

GENERAL SCIENCE GRADE 10: RATES OF REACTIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.5 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.5: Rates of Reactions
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Rates of reactions — meaning and definition of rate of a chemical reaction – Measuring rates of reactions — methods for following reactions (gas volume collected, mass lost, colour change, precipitate formation) – Factors affecting the rate of reactions: temperature – Factors affecting the rate of reactions: concentration – Factors affecting the rate of reactions: surface area – Factors affecting the rate of reactions: catalysts – Factors affecting the rate of reactions: light – Rates of reactions in biological processes (enzyme-controlled reactions, respiration, photosynthesis) – Rates of reactions in chemical and physical processes (industrial processes, food preservation, cooking) – Importance of optimum conditions in biological, chemical and physical processes
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe rate of chemical reactions; - perform experiments involving the rate of chemical reactions; - explain factors that influence the rate of reactions in biological, chemical and physical processes; - appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life.
Core Competencies	– Communication and Collaboration: as the learner discusses with peers the factors that affect the rate of chemical reactions and presents findings in plenary, including in the context of Kenyan industrial and agricultural processes. – Digital Literacy: as the learner uses digital devices and print media to research on the importance of optimum conditions in biological, chemical and physical processes (e.g., tea processing in Kericho, fish preservation at Lake Victoria). – Critical Thinking and Problem Solving: as the learner designs and interprets experiments to investigate factors affecting reaction rates, identifies variables, and draws evidence-based conclusions. – Learning to Learn: as the learner reflects on experimental outcomes, adjusts procedures, and connects findings to real-life Kenyan contexts such as cooking ugali, fermentation of uji, or rusting of iron sheets.
Core Values	– Responsibility: as the learner carefully uses digital devices to search for information on the importance of optimum conditions in industrial processes, and handles laboratory apparatus safely during reaction-rate experiments.

	– Respect: as the learner respects others' opinions while discussing with peers in a group the importance of efficient industrial processes and the role of reaction rates in daily Kenyan life. – Integrity: as the learner records experimental observations honestly and reports results accurately without falsification. – Unity: as the learner collaborates equitably with group members during practical investigations, sharing roles and responsibilities regardless of gender or background.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner frames investigable questions about why maize porridge (uji) thickens faster at higher temperatures or why food goes bad more quickly in Mombasa's heat than in Nairobi's cooler climate. – Planning and Carrying Out Investigations: as the learner designs controlled experiments to test the effect of temperature, concentration, surface area, and catalysts on reaction rates, identifying independent, dependent, and controlled variables. – Analysing and Interpreting Data: as the learner collects quantitative data (volume of gas produced, time for precipitate to form) and constructs tables and graphs to identify patterns and trends. – Constructing Explanations: as the learner uses particle/collision theory to explain why increasing temperature or concentration speeds up reactions, connecting molecular-level reasoning to observable phenomena. – Engaging in Argument from Evidence: as the learner evaluates experimental results, compares findings with peers, and defends or revises claims about which factors most significantly affect reaction rates. – Obtaining, Evaluating, and Communicating Information: as the learner researches optimum conditions in Kenyan industries (tea processing, cement manufacture, oil refining at Mombasa) and communicates findings using digital and print media.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: as the learner observes safety measures when carrying out activities to investigate rates of reactions (use of protective equipment, safe handling of dilute acids and hydrogen peroxide, proper disposal of chemical waste). – Financial Literacy: as the learner explores how controlling reaction rates in Kenyan industries (e.g., Bamburi Cement, EABL, tea factories in Kericho) reduces costs, saves energy, and improves product quality, linking chemistry to economic decision-making. – Environmental Issues: as the learner investigates how uncontrolled reaction rates contribute to environmental problems such as acid rain, corrosion of iron-sheet roofing in coastal Mombasa, and food spoilage, and explores mitigation strategies. – Socio-Economic and Environmental Education: as the learner discusses how optimising reaction rates in food processing, medicine production, and agriculture (e.g., use of fertilisers) supports Kenya's economic development and food security goals.
Career Connections	– Industrial Chemist: designing and optimising chemical processes (fertiliser production at Kenya's Athi River, cement manufacture) by controlling reaction rates for maximum efficiency and safety. – Food Scientist / Food Technologist: applying knowledge of reaction rates to food preservation, fermentation (yoghurt, uji, busaa), and value addition of Kenyan agricultural produce such as maize, milk, and cassava. – Pharmacist / Pharmaceutical Scientist: understanding reaction rates in drug formulation, shelf-life determination, and the enzymatic processes underlying medicine efficacy. – Agricultural Extension Officer: advising farmers in the Rift Valley and Central Kenya on how temperature, moisture, and soil chemistry affect nutrient-release rates from fertilisers and organic matter decomposition. – Environmental Scientist: monitoring and managing corrosion, pollution, and degradation processes influenced by reaction kinetics in Kenyan ecosystems including Lake Victoria and coastal marine environments. – Biochemist / Medical Laboratory Scientist: applying enzyme kinetics to clinical diagnostics, metabolic studies, and biotechnology research in Kenyan hospitals and research institutions such as KEMRI.
Focus for Lessons	This unit investigates the rate of chemical reactions and the factors that control how fast reactions proceed in biological, chemical, and physical systems. Learners progress from qualitative observation of everyday Kenyan phenomena — such as why ugali cooks faster on a high flame, why sukuma wiki wilts and decomposes, and why iron-sheet roofs rust faster at the Mombasa coast than in Nairobi — to quantitative experimental investigations of temperature, concentration, surface area, catalysts, and light. The unit

	employs a phenomenon-driven, inquiry-based storyline in which learners build and revise a conceptual model of particle collisions to explain and predict reaction-rate changes, culminating in an evaluation of how Kenyan industries and households can optimise conditions for safety, efficiency, and sustainability.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive? KICD KEY INQUIRY QUESTION: 1. How do various factors affect the rate of reactions?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Disappearing Thiosulfate Cross: A teacher places two identical sheets of white paper, each marked with a bold black cross (×), on a desk at the front of class. Two beakers are set up: both contain dilute hydrochloric acid (HCl), but Beaker A sits in a tray of cold water (approximately 10 °C, simulating a cool Nairobi highland morning) while Beaker B sits in a tray of warm water (approximately 50 °C, simulating afternoon heat in Mombasa or Kisumu). Sodium thiosulfate solution of identical concentration is poured simultaneously into both beakers, which are then placed over the paper crosses. Students observe both beakers from above. In Beaker B (warm), a milky-yellow precipitate of sulfur forms rapidly and the cross disappears from view within about 30–40 seconds. In Beaker A (cold), the cross remains visible for several minutes before it too eventually disappears. Students are immediately puzzled: both beakers contain exactly the same chemicals in the same amounts — so WHY does Beaker B's reaction finish so much faster? No explanation is given. The observation is recorded in students' science notebooks and the Driving Question Board is opened. This phenomenon anchors all eight lessons: every factor investigated (temperature, concentration, surface area, catalysts, light) is connected back to the question of what must be happening at the particle level inside the beaker to cause the dramatic difference in speed — and how that same principle explains fermentation of uji, rusting of iron roofing sheets, cooking of githeri, and enzyme activity in the human gut.
Storyline Thread	Lesson 1 — What Is a Rate of Reaction? (Anchoring Phenomenon Launch) Students observe the disappearing thiosulfate cross demonstration (cold vs. warm beaker). They record observations, generate initial explanations, and post questions on the Driving Question Board (DQB). The teacher introduces the formal definition of rate of reaction and discusses methods of measuring it (time for precipitate to form, volume of gas collected, loss of mass). Students make initial predictions about what caused the difference in speed between the two beakers and sketch a beginning particle model. Lesson 2 — Measuring Reaction Rates: Designing the Investigation Students learn to quantify reaction rate using the thiosulfate–HCl system (rate = 1/time for cross to disappear). They plan a controlled investigation, identifying independent variable (one factor to change), dependent variable (time/rate), and controlled variables. Groups design a data table and predict the shape of a rate-vs.-factor graph. The DQB is updated with new questions emerging from the planning process. Lesson 3 — Factor 1: Effect of Temperature on Reaction Rate (Experiment) Groups carry out the thiosulfate–HCl experiment at four temperatures (cold water ~10 °C, room temperature ~25 °C, warm ~40 °C, hot ~55 °C). They record time for the cross to disappear, calculate rate (1/time), and plot a graph of rate vs. temperature. Class data is pooled. Students construct an explanation using particle collision theory: higher temperature → particles move faster → more frequent and more energetic collisions → faster reaction. The mango-rotting and ugali-cooking supporting phenomena are revisited.

	Lesson 4 — Factor 2: Effect of Concentration on Reaction Rate (Experiment) Students investigate the effect of varying sodium thiosulfate concentration (keeping temperature and volume constant) on reaction rate. They dilute the thiosulfate to four concentrations, measure time for cross to disappear, and plot rate vs. concentration. Collision theory explanation is extended: higher concentration → more particles per unit volume → more frequent collisions → faster reaction. Students connect this to the antacid tablet supporting phenomenon and to why a stronger acid in a car battery corrodes metal faster. Lesson 5 — Factor 3: Effect of Surface Area on Reaction Rate (Experiment) Using marble chips (calcium carbonate) and dilute hydrochloric acid, groups measure the volume of CO₂ gas produced per minute using large chips, small chips, and powder. They plot volume-vs.-time curves and compare gradients (rates). Collision theory: smaller particles → greater surface area exposed → more collision sites → faster reaction. Students connect this to why maize meal (unga) dissolves into uji faster than whole maize kernels, and why the crushed antacid tablet works faster. Lesson 6 — Factor 4: Catalysts and Light (Demonstration + Research) The teacher demonstrates the liver-and-hydrogen-peroxide experiment (fresh liver, boiled liver, sand). Students observe, explain using the concept of enzymes as biological catalysts, and distinguish catalysts from reactants. The class discusses industrial catalysts (e.g., iron catalyst in the Haber process for fertiliser production, used at Kenya's Athi River). The effect of light on reaction rate is introduced via the photographic film analogy and the role of light in photosynthesis. Students use available digital or print media to research optimum conditions in one Kenyan industrial or biological process and present findings to the class. DQB is revisited and updated. Lesson 7 — Rates of Reactions in Biological, Chemical, and Physical Processes (Application and Sense-making) Students synthesise all four factors into a unified collision-theory model. They apply their model to explain: enzyme kinetics in digestion (optimum temperature/pH), fermentation of uji and yoghurt (temperature control), rusting of iron-sheet roofs in coastal Mombasa (high humidity and salt concentration), and food preservation strategies (refrigeration, salting, drying of fish at Lake Victoria). Groups create annotated diagrams linking each Kenyan context to the relevant factor(s) affecting reaction rate. The DQB is reviewed: which original questions have been answered? Which remain? Lesson 8 — Model Building, Assessment, and Community Connection (Synthesis Lesson) Students revise and finalise their cumulative particle-collision model, incorporating all four factors and both biological and chemical contexts. Groups present their models and explain how manipulating reaction-rate factors can solve a real Kenyan problem of their choice (food spoilage, industrial efficiency, medical dosage, environmental corrosion). A brief written assessment (predict–explain task using a novel Kenyan scenario) is administered. The DQB is closed out: students vote on the most satisfying answer to the driving question and identify remaining curiosities for further study. The lesson closes by connecting to careers (food scientist, industrial chemist, pharmacist, KEMRI biochemist) and to the KICD values of responsibility and respect in scientific work.
Supporting Phenomena	– Supporting Phenomenon 1 — The Rotting Mango Problem: A market trader in Kisumu notices that mangoes left in the open stall at 35 °C go soft and mouldy within two days, while identical mangoes kept in a cool box at 8 °C stay fresh for over a week. Students are asked: is this a chemical/biological reaction? What factor is changing? How does this connect to the disappearing cross? – Supporting Phenomenon 2 — Faster Fizz with Crushed vs. Whole Antacid Tablets: A student at a Nairobi chemist is given an antacid tablet for stomach acid (excess HCl). She notices the crushed powder fizzes and relieves pain far faster than the whole tablet. Why does breaking the tablet into smaller pieces speed up the reaction between the antacid (calcium carbonate) and stomach acid?

- Supporting Phenomenon 3 — Tea Factory Catalyst Mystery: Workers at a Kericho tea factory add small amounts of a special enzyme (oxidase) to speed up the oxidation of tea leaves during processing. Without the enzyme, oxidation takes hours; with it, the same reaction completes in minutes — yet the enzyme is recovered unchanged at the end. How can something speed up a reaction without being used up?
- Supporting Phenomenon 4 — Hydrogen Peroxide and the Liver: A biology teacher demonstrates that adding a small piece of fresh cow's liver (obtained from a Nairobi butchery) to hydrogen peroxide produces a dramatic burst of oxygen bubbles, while adding sand produces nothing. Boiled liver produces almost no bubbles. Students must explain what the liver contains, why heat destroys the effect, and how this connects to reaction rates in the human body.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute lessons)): What Is a Rate of Reaction? Launching the Disappearing Cross

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Launch the anchoring phenomenon of the Disappearing Cross. Learners define rate of reaction and meet three ways of measuring it, then raise the questions that will drive the next seven lessons.
Knowledge	By the end of the Sub Strand, the learner should be able to: a) describe rate of chemical reactions. In this lesson learners state that rate of reaction is how fast reactants are used up or products are formed. They name three ways to measure it: time for a change (such as a precipitate hiding a cross), volume of gas produced, and loss of mass. They recognise that the same chemicals and amounts can react at different rates.
Skills	Predicting, observing and recording in the science notebook, asking investigable questions, using particle ideas to build an initial model, communicating ideas in pairs and in plenary (NGSS practices: Asking Questions, Developing and Using Models, Constructing Explanations).
Attitudes	Curiosity about everyday changes (a mango rotting, ugali cooking); Respect, as learners listen to others' opinions when sharing predictions; Responsibility, as learners follow safety rules in the laboratory. Core competency: Communication and collaboration, as learners discuss the meaning of the rate of a chemical reaction with peers.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	This is the launch lesson. It shows the Disappearing Cross once and opens the Driving Question Board (DQB) with the unit question: Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive? Lesson 2 will turn the times into rates and plan a fair test.
Safety Notes	Sulfur dioxide is an irritant, so the teacher runs the demo in a ventilated room using small quantities, with goggles worn by the teacher and by learners within 2 m. Rinse both beakers straight after the cross vanishes. Use dilute acid only. Do not let learners smell the beakers or touch the acid. Hot water is about 50 C, so handle the warm bath with care. Wash hands after the lesson.

B. LESSON OVERVIEW

Mwalimu Achieng's Grade 10 class in Nakuru County begins with everyday changes they already know: a mango rotting in a hot kiosk and ugali cooking on a jiko. Learners predict what will happen when sodium thiosulfate is poured into hydrochloric acid in a cold beaker (Beaker A, 10 C, a cool Nairobi highland morning) and a warm beaker (Beaker B, 50 C, an afternoon in Mombasa or Kisumu), each placed over an identical black cross on white paper. The teacher runs the demo once. The cross vanishes at 35 s in Beaker B and at 270 s in Beaker A, so B is 7.7 times faster (270/35). Both beakers had the same chemicals in the same amounts and only temperature differed. Learners record what they see, define rate of reaction and meet three ways of measuring it. They then sketch first particle ideas, and the class opens the DQB with the questions they cannot yet answer. No explanation of the cause is given in this lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners start with a 3-minute	Greet the class: 'Today we begin	Eliciting prior knowledge through	Check notebooks for: (a) a written

 VIDEO: Introduction to reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Predicting Precipitates (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	quick-write in their science notebooks: 'Think of something that changes fast (for example, ugali cooking on a hot jiko) and something that changes slowly (a mango rotting). What makes one fast and one slow?' They then see the setup only: two identical white sheets with a bold black cross, Beaker A in a cold-water tray (about 10 C, a cool Nairobi highland morning) and Beaker B in a warm-water tray (about 50 C, an afternoon in Mombasa or Kisumu). Each beaker holds 10 cm³ of 1.0 mol/dm³ HCl. 50 cm³ of 0.15 mol/dm³ sodium thiosulfate will be added to both at the same moment. Learners predict in writing: (1) What will you see in each beaker? (2) Will the two beakers behave the same or differently? (3) Draw what you think will happen to the cross over time.	with a puzzle from our own Kenyan weather.' Ask: 'Why does a mango rot faster in a hot kiosk than in a cool house?' WAIT TIME: 10-15 seconds in silence before taking answers. Cold-call 3 learners for their fast/slow examples and write them on the board in two columns. Point to the setup without saying what will happen: 'Both beakers hold exactly the same chemicals in the same amounts. Only the temperature of the water bath is different.' Say: 'Commit to a prediction in your notebook. You will not be marked wrong, because predictions are how scientists begin.' Walk around and read 4-5 predictions without correcting them. Choose 2 contrasting predictions to share, for example 'both the same' and 'the warm one is faster'.	individual prediction (Predict-Observe-Explain). Writing a prediction before seeing evidence makes learners' everyday ideas about fast and slow changes visible, so they can later compare what they expected with what they observed, which builds understanding through conflict and confirmation.	prediction that names a difference or no difference between A and B, (b) at least one everyday example of a fast or slow change, (c) a sketch of the cross. Tally on the board how many predict 'B faster', 'A faster' and 'same'. Note learners who give a reason (for example 'heat makes it faster') as ready for deeper questions, and learners who only describe as needing prompting.
Observe Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Predicting Precipitates Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners gather around the front bench, staying at least 2 m back, and view both beakers from above. Mwalimu Achieng pours the sodium thiosulfate into both beakers at the same moment and places them over the identical crosses. Learners watch the milky-yellow precipitate form. Using a classroom clock or a stopwatch app, each pair records: time the cross vanishes in B, time the cross vanishes in A, and a description of the colour changes (clear to milky yellow). Results: Beaker B at 35 s, Beaker A at 270 s (4 min 30 s). Learners calculate B is faster by $270 - 35 = 235$ s, and $270/35 = 7.7$ times. The yellow solid is named as sulfur. They then look at three photographs or drawings of rate measurement and match each to a	Before pouring say: 'Eyes on the cross. Pairs, one of you watches Beaker A, the other Beaker B, and call out when no part of the black lines can be seen from above.' Announce the start time, pour both beakers together and call out the time. Do NOT explain the difference. After the cross in B vanishes, say: 'Stop. What did you see?' WAIT TIME: 10-15 seconds. Cold-call 3 learners to describe only what they saw, with no reasons yet. Ask: 'Were the chemicals and amounts in the two beakers the same?' Cold-call 2 learners. Rinse both beakers straight away and tell learners the reason: 'sulfur dioxide gas is an irritant'. Introduce the three measuring methods with the sentence: 'The cross method	Structured observation with data recording and claim-evidence separation. Learners write observations (what was seen and the times) separately from ideas (why), which keeps evidence distinct from explanation. Linking the definition to three measurement methods gives them a shared vocabulary, so the phenomenon can be described quantitatively with 'rate'.	Look for: times recorded accurately (35 s and 270 s) with units; the difference 235 s and the ratio 7.7 calculated correctly; the yellow solid named as sulfur; the statement that the chemicals and amounts were the same in both beakers. Ask 2 learners to match each measuring method to its example. Indicator of success: a learner can say 'rate is how fast something is used up or made' and give the cross as a time-for-a-change example.

	method: (1) time for a change, for example the cross disappearing; (2) volume of gas collected in a syringe or measuring cylinder; (3) loss of mass on a balance as gas escapes. Learners copy the definition: 'Rate of reaction is how fast reactants are used up or products formed.'	measures the time for a change. What else could we measure if a gas were given off?' Take 3 ideas and name volume of gas and loss of mass. Do not give the answer to why B was faster.		
Explain Phase  VIDEO: Introduction to reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Predicting Precipitates Using Solubility Rules (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	In pairs, learners use the sentence frame 'Beaker B reacted faster than A because _____' and write one idea, then draw a particle sketch of what could be happening inside each beaker (dots for particles with arrows for movement). Otieno suggests that everything in the beaker is made of tiny particles that may be moving differently in warm and cold water. Learners test this idea by drawing the particles in A and B and adding a note about what might happen when particles meet. They also answer: 'Do you think both beakers will end with the same amount of sulfur? Why or why not?' Then they link back to their own examples (mango rotting, ugali cooking) and write one sentence saying whether those changes also have a 'rate'. Each pair writes a 'we think / we wonder' statement: one thing they think is happening and one thing they cannot explain yet.	Say: 'Turn to your partner. You have 2 minutes to write why B might be faster. You do not need the right answer, only your best idea.' Circulate and look for particle drawings. WAIT TIME: 10-15 seconds after asking 'What do you think the particles are doing in each beaker?' Invite Otieno to share his particle idea, then cold-call 3 more pairs to add to it or challenge it. Record the ideas on the board as 'working ideas' and do not mark them correct or incorrect: 'Hold on to that idea, we will test it.' Ask: 'Is the warm reaction making MORE sulfur, or the same amount in less time?' Take 2 answers and say that this will be checked later. Redirect any claim that heat itself 'is' the rate: 'Rate means how fast, so what do we measure to find it?' Do not teach collision theory now, since it is built in Lesson 3.	Constructing initial explanations with an emergent particle model. Learners sketch and write ideas in pairs before whole-class talk, so they commit to a model they can revise. Treating explanations as working ideas helps them see that science explanations are revised using evidence.	Listen and look for: particle drawings that show particles in both beakers; any link between temperature and particle motion; a statement about whether the final amount of sulfur is the same or different; correct use of 'rate' for how fast a reaction happens. Record who needs support with the idea that rate is not the same as amount made. A pair meets the target when it can state one idea and one question about the phenomenon.
Driving Question Board (DQB) Creation  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES	Wanjiru asks the first 'why' question aloud: 'Why did the same chemicals react so differently?' Learners write their own questions on sticky notes, one per note. Examples include 'Would a bigger piece of marble react faster?' and 'Why does ugali cook faster on a hot jiko?' Amina asks about iron roofing sheets rusting in Mombasa	Display the Driving Question: 'Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive?' Say: 'What questions do YOU have? Write one per sticky note, starting with how, why or what if.' WAIT TIME: 10-15	Driving Question Board with question clustering. Learners' own questions set the direction for the unit, and clustering shows that different questions share a common cause, so each later lesson can be tied back to the Disappearing Cross.	Check that each learner has posted at least one question and that notes are specific rather than vague (for example, 'Does a stronger acid change the time?'). Count how many questions refer to a possible factor (temperature, concentration, size, catalyst, light) and how many to Kenyan contexts. Success indicator: a

for similar videos EXERCISE: Predicting Precipitates Using Solubility Rules Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	and Kiprono asks about uji fermenting in Kericho. Learners post the notes on the DQB at the back wall under the unit question, then work in groups to cluster questions into categories such as temperature, amount or strength of acid, size of pieces, helpers (catalysts), and real-life uses. Each group chooses one question they most want to investigate.	seconds of silent thinking before writing. Call on Wanjiru to read the first question and stick it at the centre of the board. Cold-call 4 learners, including Amina and Kiprono, to read and place their notes. Say: 'Which questions look alike? Group them.' Help the class label the clusters. Point out that some questions can be answered by an experiment and some are about real-life uses. Tell learners that the board stays up and that they will move notes to 'answered' as the lessons proceed.		learner can sort a question into a cluster and say why it belongs there.
Model Building Phase VIDEO: The kinetic molecular theory of gases <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Predicting Precipitates (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Each learner draws an Initial Model in the science notebook with three parts: (1) a labelled sketch of the two beakers and crosses; (2) a particle picture of what might be happening inside A and B; (3) a caption in words: 'The rate of reaction is... Beaker B was faster because I think...'. Learners then complete the summary table prompt: What we observed, What we learned, What we can explain so far. They finish by naming the three ways of measuring rate and giving one Kenyan example for each, for example a stopwatch for a cross, a syringe for gas from an antacid tablet, or a balance for mass loss.	Say: 'This is your first model. It will change, and that is the point. Include only ideas you actually believe right now.' Show a blank example frame on the board without any answer. WAIT TIME: 10-15 seconds before learners begin drawing. Circulate and ask 5 learners: 'What are the particles doing in your warm beaker?' and 'How would you measure the rate if gas were produced?' Cold-call 3 learners to share their model captions. Close by pointing to the DQB: 'Next lesson we turn the two times into rates and plan a fair test. Keep your model safe.' Collect the notebooks or have learners paste the model in the front pages.	Initial model building and revision. Learners externalise their thinking in a drawing and a caption, which becomes a baseline. Later lessons revise the same model, so learners see their own understanding grow.	Check each model for: both beakers shown; correct times 35 s and 270 s; a particle picture; a correct definition of rate; three measurement methods named (time for a change, volume of gas, loss of mass). Level guide: Meets expectations if the definition and at least two methods are correct; Approaches if only one is correct. Use the models to group learners for support in Lesson 2.

D. TEACHER REFLECTION

After the lesson, consider: Did every learner see the cross vanish in both beakers, and could they all record 35 s and 270 s? Which predictions surprised you, and which everyday examples (mango, ugali, uji) did learners offer? Did any learners confuse 'rate' with 'amount of product'? Did Otieno's particle idea and Wanjiru's first question help to open the DQB for the whole class? Which learners were quiet and need a different way to contribute next time? Were the safety steps followed: ventilation,

goggles, immediate rinsing of the beakers? What would you change about pairing or timing for Lesson 2, when the times are converted to rates?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the demo, the cross vanished at 35 s in warm Beaker B (50 C) and at 270 s in cold Beaker A (10 C). A yellow precipitate of sulfur formed in both. The chemicals and amounts were the same and only temperature differed, so B was 7.7 times faster.
What did I learn?	Rate of reaction is how fast reactants are used up or products are formed. It can be measured by the time for a change (such as a precipitate hiding a cross), by the volume of gas produced, or by loss of mass.
How does this explain the phenomenon?	So far we can only say that temperature seems linked to speed and that particles may be involved. We have a first particle sketch and DQB questions, but we cannot yet explain why the warm reaction was faster.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Measuring Reaction Rates: Designing a Fair Test

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students turn the demo times (35 s and 270 s) into rates using $\text{rate} = 1/t$ and plan a fair test for the disappearing cross by naming the independent, dependent and control variables. This prepares them to investigate temperature in Lesson 3.
Knowledge	By the end of the lesson the learner should be able to: (a) describe rate of chemical reactions (KICD SLO a) as change in amount of reactant or product per unit time; (b) use $\text{rate} = 1/t$ for the thiosulfate cross method and quote rate in $\times 10^{-3} \text{ s}^{-1}$; (c) identify the independent, dependent and control variables in a rate investigation.
Skills	Science and Engineering Practices: Planning and Carrying Out Investigations (identify and control variables), Analysing and Interpreting Data (convert times to rates), Using Mathematics and Computational Thinking (calculate $1/t$ and the ratio of rates), Asking Questions.
Attitudes	Core competency: Communication and collaboration as learners discuss with peers the factors that affect the rate of chemical reactions. Value: Respect, as learners listen to others' opinions when critiquing test plans. PCI: Safety and security, as learners observe safety measures when planning and carrying out activities to investigate rates of reactions.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lesson 1 launched the disappearing cross demo (Beaker A at 10 C: 270 s; Beaker B at 50 C: 35 s) and defined rate of reaction, and the DQB holds the question of why B was faster. Lesson 2 gives students the tools to compare reactions fairly. They convert times to rates and name the variables so that in Lesson 3 they can test temperature. No new results are introduced here. Explaining WHY B is faster is left for Lesson 3 and collision theory.
Safety Notes	Sulfur dioxide (SO_2) is an irritant. Only the teacher repeats the small-scale demo, with 50 cm ³ thiosulfate and 10 cm ³ HCl, in a ventilated room. Everyone wears goggles. Rinse beakers straight after the cross vanishes. Keep hot-water baths away from table edges. Do not smell the beakers. Report spills to the teacher.

B. LESSON OVERVIEW

Mwalimu Achieng reopens Lesson 1: Beaker B cleared the cross at 35 s and Beaker A at 270 s. Wanjiru's DQB question, 'Why was B faster?', cannot be answered until we can compare reactions fairly and numerically. Students predict how to turn times into a single number for speed, then calculate rates from the demo times ($1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$; $1/270 = 3.7 \times 10^{-3} \text{ s}^{-1}$). They observe the demo set-up again by examining a quantities card and sort what was the same in both beakers (50 cm³ of 0.15 mol/dm³ thiosulfate, 10 cm³ of 1.0 mol/dm³ HCl, identical crosses, same observer, same beaker type) from what differed (temperature). In the Explain phase they name the independent, dependent and control variables and critique a flawed test plan. They add questions to the DQB and draft a first model note on what a fair test must keep the same. They do not yet explain why the warm beaker is faster; that is Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students look at the Lesson 1 notebook record: Beaker A (10 C)	Write on the board: 'Beaker B = 35 s, Beaker A = 270 s.' Say:	Prediction-before-instruction with pair-share: students commit to	Look for: (1) writes a method in the notebook before talking; (2) can

VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Independent versus dependent variables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	= 270 s, Beaker B (50 C) = 35 s. Mwalimu Achieng asks, 'Time is smaller for the faster reaction. If I want one number where bigger means faster, what could I do with the time?' Individually (2 minutes) they write a prediction in their science notebook: a way to turn 35 s and 270 s into numbers where the faster reaction gets the bigger number. They also predict how many times faster B is, using their own method. Then they pair-share with a neighbour (Kiprono might suggest 'flip the time' based on his tea-brewing timing at home in Kericho).	'Yesterday we said B was faster. Today we want a number for HOW fast.' Ask: 'If a reaction takes a SHORTER time, what should happen to its speed number?' WAIT TIME 10-15 seconds in silence. Cold-call 3 students (not volunteers) to share their method. Do not confirm or correct. Say: 'Keep your prediction; we will test it with the demo numbers.' Circulate and note who divides, who subtracts and who uses time directly.	their own idea of how time relates to speed, which exposes the common misconception that a bigger time number means a bigger rate. Revisiting this prediction at the end of the Explain phase lets them see their thinking change.	state that shorter time must give a bigger rate number; (3) notes any 'subtract the times' answers as a misconception to revisit. Tally on a class list: number using 1/t, number using subtraction, number with no method.
Observe Phase VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Independent versus dependent variables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Students gather evidence in two parts. Part 1 (data): Mwalimu Achieng shows the demo quantities card on the board: 50 cm³ of 0.15 mol/dm³ sodium thiosulfate plus 10 cm³ of 1.0 mol/dm³ HCl in each beaker; Beaker A 10 C; Beaker B 50 C; identical black crosses on white paper; cross 'gone' when no part of the black lines can be seen from above. In groups of four they calculate the rate for each beaker with rate = 1/t: B = 1/35 = 0.0286 s⁻¹ = 28.6 x10⁻³ s⁻¹; A = 1/270 = 0.0037 s⁻¹ = 3.7 x10⁻³ s⁻¹. They compare with their predictions. Part 2 (observation): the teacher quickly repeats a small-scale cross test in one beaker at room temperature, and students use a stopwatch to time when the cross is 'gone', noticing that different observers disagree by a few seconds. They record the problem in their notebooks.	Distribute calculators (or let students use paper). Say: 'Use only the numbers on the card. Work out 1 divided by the time in seconds for each beaker.' Walk around and ask: 'Which beaker has the bigger rate number? Does that match what you saw?' Ask: 'How many times bigger is 28.6 than 3.7?' (about 7.7, the same ratio as 270/35). WAIT TIME 10-15 seconds after each question. For the repeat demo, wear goggles, work in a ventilated room, use the small quantities and rinse the beaker at once. Ask 4 different students to say the second at which THEY judge the cross gone, and write all 4 times on the board. Ask: 'Why do our four numbers differ?' Cold-call 3 students. Do not introduce collision theory.	Data transformation and noticing: converting raw times into rates shows that the same evidence can be re-expressed so that 'faster' means 'bigger number'. The disagreement between observers makes a need for agreed rules (what counts as 'gone') visible, building the idea of control before the term is introduced.	Observable indicators: (1) correct calculation of 28.6 and 3.7 x10⁻³ s⁻¹ in notebooks; (2) uses the unit s⁻¹ or x10⁻³ s⁻¹; (3) can state B's rate is about 7.7 times A's; (4) suggests that a fixed observer or rule is needed when the four times differ. Check 5 notebooks while walking round.
Explain Phase VIDEO:	Students make sense of the evidence. Mwalimu Achieng introduces the words independent	Say: 'Everyone stand and point to the one thing that was different between Beaker A and Beaker B.'	Claim-Evidence-Reasoning with error analysis: students must justify each control variable by	Observable indicators: (1) correctly sorts temperature as independent, time or rate as dependent, and at

Dependent & independent variables: graphing <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Making appropriate scatter plots <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	variable (the factor you change), dependent variable (what you measure: the time, or the rate) and control variables (everything else kept the same). Groups return to the quantities card and make a three-column table for the anchor demo: independent = temperature (10 C vs 50 C); dependent = time for the cross to disappear (converted to rate = $1/t$); controls = 50 cm³ of 0.15 mol/dm³ thiosulfate, 10 cm³ of 1.0 mol/dm³ HCl, same type of beaker, identical black crosses, same observer, same lighting, the rule for when the cross is gone, simultaneous pouring. Then they critique a flawed plan from 'Otieno's cousin' who used a big beaker with 80 cm³ of thiosulfate for the warm test and a small beaker with 50 cm³ for the cold test, and changed observers. They list at least three flaws and rewrite the plan. Finally they state clearly that rate here is a relative rate, since $1/t$ is not an absolute measure of amount per second. They also note that both beakers end with the same amount of sulfur; the rate says how fast, not how much.	Then ask: 'What name do scientists give to the thing we change on purpose?' Cold-call 3 students for each of the three variable names. WAIT TIME 10-15 seconds before calling. Prompt: 'Why must the 50 cm³ and 10 cm³ be the same in both beakers? What would happen to our conclusion if they were not?' Then hand out the flawed plan; give 6 minutes in groups. Cold-call 4 groups to report one flaw each. Press: 'Which flaw would make us wrongly blame temperature?' Reinforce: 'We say rate, never rate of heat.' Revisit the Predict phase: 'Who changed their idea about subtracting times? What changed it?' Do not explain why heat speeds up the reaction; say: 'That question is for Lesson 3 and stays on the DQB.'	showing how breaking it would weaken the claim that temperature caused the difference. Critiquing a flawed plan builds understanding of why fair tests matter more than simply following a recipe.	least four controls; (2) identifies at least 3 of the flaws (different volume, different beaker, different observer); (3) explains in a sentence why a change in two things at once makes the result unclear; (4) uses the word 'rate' not 'speed of heat'. Use a thumbs-up/sideways/down check on 'I can name the three kinds of variable', and note who shows sideways or down for a small-group follow-up.
Driving Question Board (DQB) Creation VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Analyze	Students return to the DQB on the back wall. Each writes on sticky notes (1) one new question about how to measure or compare rates fairly, and (2) one question about what is happening inside the beaker. Examples might include 'Can we use $1/t$ for other reactions, like rusting of roofing sheets in Mombasa?' (inspired by Amina) or 'How would we measure the rate of uji fermenting?' (inspired by Kiprono). Wanjiru's earlier 'why' question is revisited and moved into a 'Needs the particle idea'	Say: 'Write one question you could answer with a fair test and one you could not yet answer.' Give 3 minutes of silent writing. Cold-call 5 students to read and place their notes. Ask: 'Which questions can we test with the cross method? Which need a different way of measuring, like collecting gas or loss of mass?' WAIT TIME 10-15 seconds. Mark questions answered today with a green dot (for example, 'How do we get a number for speed?') and leave the 'why' questions unmarked. Say:	Driving Question Board sorting: grouping questions by type (measure, factor, mechanism) shows students what can already be answered and gives a map of what evidence is needed next. It keeps student curiosity at the centre of the sequence.	Observable indicators: (1) each student produces two notes; (2) at least half of the notes refer to the cross demo or rate; (3) students correctly place notes in categories and justify a placement; (4) at least two students link a Kenyan context (rusting, uji, githeri) to measuring rate. Record question quality: factual, testable or mechanism-based.

relationships between variables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	column, and the class sorts questions into three groups: How do we measure rate? What factors change rate? Why do those factors work?	'These are your questions; we answer them with evidence, not by guessing.'		
Model Building Phase VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Independent versus dependent variables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Students begin the 'My Rates Model' page in their science notebooks. They (1) draw the two beakers, label 10 C and 50 C, and write the times and rates beneath each (270 s, $3.7 \times 10^{-3} \text{ s}^{-1}$; 35 s, $28.6 \times 10^{-3} \text{ s}^{-1}$); (2) add a box titled 'Fair test checklist' listing the independent variable, dependent variable and controls; (3) sketch a first particle idea of what could be different inside the warm and cold beakers (a beginning guess only, using dots for particles), marking it clearly with a question mark because the explanation is still to be tested in Lesson 3. They write a two-sentence summary beginning, 'To compare how fast reactions are, I...'	Say: 'Your model today has two parts: a measuring part, which you can finish now, and a particle part, which is only a guess.' Circulate and ask: 'What does your picture show that is different between the beakers? What is the same?' Cold-call 3 students to share their sketches with the document camera or by holding up notebooks. WAIT TIME 10-15 seconds before each call. Prompt: 'Both beakers end with the same amount of sulfur. How might your sketch show that?' Do not correct particle sketches; accept all and record common ideas, such as 'particles move faster' (Otieno's idea from Lesson 1), for Lesson 3 testing. Close with the exit question below.	Cumulative model building with revisable drafts: students record what they can already support with evidence (rates, fair test) and label what remains a hypothesis (particle behaviour). Separating the two helps them tell evidence from speculation.	Observable indicators: (1) both rates are correctly written with units on the diagram; (2) the fair test checklist includes temperature as independent and at least three controls with their values (50 cm ³ , 10 cm ³); (3) the particle sketch shows some difference between warm and cold and is marked as tentative; (4) the exit slip answers 'Why is it unfair to compare a cold test in a large beaker with a warm test in a small beaker?' Collect exit slips and sort into secure / partly secure / not yet.

D. TEACHER REFLECTION

Before Lesson 3, check: Did students use $1/t$ correctly and see that a shorter time means a bigger rate? Could they explain why each control variable matters, not just list it? Which students still subtract times or confuse the dependent variable with the rate? Did the four different observer times spark a real conversation about what 'gone' means? Did any student move on to explaining why B was faster, and did I hold the mechanism for Lesson 3 as planned? Which DQB questions should shape the Lesson 3 temperature investigation (10, 25, 40 and 55 C)? Make sure goggles were worn and the room was ventilated during the repeat demo.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We calculated the rate for the demo: Beaker B (50 C, 35 s) has rate $1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$ and Beaker A (10 C, 270 s) has rate $1/270 = 3.7 \times 10^{-3} \text{ s}^{-1}$, so B is about 7.7 times faster. Different observers judged the cross gone at slightly different times.
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What did I learn?	Rate = $1/t$ for the cross method, so a shorter time gives a bigger rate. In a fair test we change only the independent variable (temperature), measure the dependent variable (time, then rate) and keep control variables the same (50 cm ³ thiosulfate, 10 cm ³ HCl, beaker, cross, observer).
How does this explain the phenomenon?	We can compare the two beakers fairly because the chemicals, amounts and way of judging the cross were the same and only temperature differed. We cannot yet explain WHY the warm beaker reacts faster; that question stays on the DQB for Lesson 3.

LESSON 3 (80 minutes (double lesson of 2 x 40 minutes)): Factor 1: Temperature

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how temperature affects the rate of the sodium thiosulfate-hydrochloric acid reaction. They use the class data table to build the first statement of collision theory, and use it to explain why warm Beaker B (35 s) was 7.7 times faster than cold Beaker A (270 s).
Knowledge	By the end of the lesson the learner should be able to: (a) describe how temperature changes the rate of the thiosulfate-HCl reaction using the class table (10 C = 270 s; 25 C = 110 s; 40 C = 52 s; 55 C = 26 s); (b) calculate rate = $1/t$ in $\times 10^{-3} \text{ s}^{-1}$ (3.7, 9.1, 19.2, 38.5); (c) explain, using particles, kinetic energy, effective collisions and activation energy, why a higher temperature increases the rate; (d) state that temperature changes the speed of the reaction and not the final amount of sulfur made.
Skills	Predicting from a model; carrying out or analysing a fair test (independent variable = temperature, dependent variable = time, control variables = 50 cm ³ of 0.15 mol/dm ³ thiosulfate and 10 cm ³ of 1.0 mol/dm ³ HCl, same observer, same lighting, same beaker type); converting time to rate; plotting rate (y-axis) against temperature (x-axis); NGSS practices: Developing and Using Models, Analyzing and Interpreting Data, Constructing Explanations.
Attitudes	Appreciate the importance of manipulating factors that affect the rate of chemical and biological processes in day-to-day life. Core competency: Communication and collaboration as learners discuss the factors that affect the rate of reactions. Values: Respect (respecting others' opinions in group discussion) and Responsibility. PCI: Safety and security, as learners observe safety measures when carrying out activities to investigate rates of reactions.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lesson 1 launched the disappearing cross: Beaker B (50 C) lost its cross at 35 s and Beaker A (10 C) at 270 s, and the class defined rate and posted 'why' questions on the DQB. Lesson 2 taught turning time into rate (rate = $1/t$) and planning a fair test with named independent, dependent and control variables. Lesson 3 uses that planning to gather evidence on the first factor, temperature. It gives the first statement of collision theory, which Lessons 4 to 6 reuse for concentration, surface area and catalysts.
Safety Notes	Sulfur dioxide (SO ₂) is an irritant. Use only the stated small quantities, work in a ventilated room, and wear goggles throughout. Rinse each beaker straight after the cross vanishes. Hot water baths at 55 C can scald, so use a thermometer, never boiling water, and carry baths with two hands. Keep asthmatic learners at the back and let them record data instead of handling the beakers. Mop up spills at once.

B. LESSON OVERVIEW

Mwalimu Achieng reopens Wanjiru's first 'why' question on the DQB: both beakers held the same chemicals in the same amounts, so why did warm Beaker B finish so much faster? Learners predict what will happen at four temperatures, then run or analyse the fair test from Lesson 2. They use 50 cm³ of 0.15 mol/dm³ sodium thiosulfate and 10 cm³ of 1.0 mol/dm³ HCl at 10, 25, 40 and 55 C. Class-mean results are 270, 110, 52 and 26 s, giving rates of 3.7, 9.1, 19.2 and 38.5 $\times 10^{-3} \text{ s}^{-1}$. In the Explain phase learners use particle drawings and a Kenyan kiosk and jiko analogy (mango rotting faster in a hot kiosk, githeri cooking faster on a hot jiko) to build the first statement of collision theory. At higher temperature particles have more kinetic energy, collide more often and more energetically, and a larger fraction of collisions exceed the activation energy, so more collisions are effective. Learners place the demo's 35 s at 50 C between the 40 C (52 s) and 55 C (26 s) rows as a check on the pattern. They add new questions to the DQB and revise their first model, noting that temperature changes the speed of the reaction and not the final amount of sulfur.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to kinetics Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Learners open their science notebooks to the Lesson 1 demo record (Beaker A 10 C, 270 s; Beaker B 50 C, 35 s). Wanjiru's DQB question is read aloud. In pairs, learners predict the time for the cross to disappear at 10, 25, 40 and 55 C and sketch a quick prediction graph of time against temperature. They also write one sentence beginning 'I think the warm one is faster because...'. Kiprono links the idea to uji and tea, and Amina to mango rotting faster in a hot kiosk in Mombasa than in a cool Nairobi home. Learners record whether they expect the same amount of sulfur at the end in all four beakers.	Mwalimu Achieng points to the DQB: "Wanjiru asked why Beaker B won't wait for Beaker A. Today we collect evidence across more temperatures, not just two." She writes the four temperatures on the board and says: "In your pairs, predict the time for the cross to vanish at each temperature. You have 3 minutes. Write numbers, not just 'faster'." She gives WAIT TIME of 10-15 seconds after asking: "Will the warmest beaker make MORE sulfur in the end, or the same amount? Think first." She cold-calls 4 pairs (one per temperature) for their predictions and records them in a class column without judging. She asks: "Where have you seen heat speed up or slow down a change at home? Mango in a kiosk? Githeri on the jiko?" and takes 3 more cold-calls. She asks learners to say what they think is happening inside the beaker, and does not reveal the answer.	Eliciting prior ideas through a Predict-Write-Share routine. Writing a prediction with a reason before any data makes learners commit to an idea, so they can later compare it with evidence and notice where their thinking changed. Linking to mango and githeri makes everyday ideas of 'hot means fast' explicit so the data can test them.	Observe whether predictions show times falling as temperature rises (look for a decreasing time trend on sketched graphs). Note which learners predict a different final amount of sulfur and which say 'the same'. Listen for particle language such as 'moving faster' versus vague 'heat helps'. Tally on a clipboard how many pairs predict an ordering of $270\text{ s} > 110\text{ s} > 52\text{ s} > 26\text{ s}$.
Observe Phase  VIDEO: Introduction to kinetics - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12	In groups of four with assigned roles (timer, pourer, observer-recorder, safety monitor), learners carry out the fair test planned in Lesson 2, or, where apparatus is limited, run one demonstrated temperature and analyse the rest from the class data. The controls are 50 cm ³ of 0.15 mol/dm ³ sodium thiosulfate and 10 cm ³ of 1.0 mol/dm ³ HCl, pre-equilibrated at 10, 25, 40 and 55 C. The same observer looks down from above at the same black cross under the same bench lighting. They time how long it takes for the cross to	Mwalimu Achieng reminds groups: "Goggles on. Small quantities. Rinse the moment your cross vanishes because sulfur dioxide is an irritant." She circulates and asks the control-variable question: "What stays the same in your beaker? Name three controls." She prompts: "Cross 'gone' means no part of the black lines is visible from above, and the same person judges every time." When groups finish she builds the table on the board and cold-calls 4 groups, one per temperature, for their times. She says: "Our class means are	Data table construction and pattern noticing with a structured 'I notice / I wonder' routine. Building the class table from learners' own measurements, then converting to rate, shows that the pattern is real and repeatable. Linking the 10 C row to Beaker A, and then noticing that each temperature step increases the rate, moves learners from a single surprising event to a general relationship.	Check that groups name independent (temperature), dependent (time) and control variables correctly. Check rate calculations: $1/270 = 3.7$, $1/110 = 9.1$, $1/52 = 19.2$ and $1/26 = 38.5$ ($\times 10^{-3}\text{ s}^{-1}$). Look for learners who read the cross consistently as 'gone' and who rinse beakers promptly. Listen for statements like 'higher temperature gives shorter time and higher rate'. Note who observes that the same amount of sulfur forms.

Search ARES for similar readings	disappear, rinse the beaker at once, and record their group time. The class pools its results into the mean table: 10 C = 270 s, 25 C = 110 s, 40 C = 52 s, 55 C = 26 s. Learners calculate rate = $1/t$ for each row: 3.7, 9.1, 19.2 and 38.5 $\times 10^{-3} \text{ s}^{-1}$. They notice that the 10 C row matches Beaker A from the demo, and that each 15 C rise roughly doubles the rate. They also check that the sulfur at the end looks the same amount in every beaker.	270, 110, 52 and 26 seconds. Convert each to a rate using 1 divided by t. Check with your partner." After posting rates she asks: "What do you notice?" and allows WAIT TIME of 10-15 seconds before taking answers from 3 learners. She asks: "How does your 10 C row compare to Beaker A in Lesson 1?" and then: "Is the amount of yellow sulfur left at the bottom different between warm and cold beakers once the reaction finishes?" She notes that rates are relative rates, quoted in $\times 10^{-3} \text{ s}^{-1}$.		
Explain Phase  VIDEO: Collision theory Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Factors Affecting Reaction Rate Practice Source: CK-12 Search ARES for similar readings	Learners plot rate (y-axis) against temperature (x-axis) from the class table and describe the shape: the rate rises steadily as temperature rises. They use the graph to place the demo's 35 s at 50 C (rate $1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$) between the 40 C row (52 s) and the 55 C row (26 s), and they check that it fits. Then, in groups, learners draw two particle diagrams of the beaker contents, one cold (10 C) and one warm (55 C). They use arrows for particle speed and show collisions, including one that is effective and one that is not. Otieno's particle idea from Lesson 1 is revisited and extended. They write a three-part explanation: warm particles have more kinetic energy, so they collide more often, and they collide with more energy, so a larger fraction of collisions reach the activation energy (the minimum energy needed for a collision to react) and are effective collisions. They link this to a mango rotting faster in a hot kiosk and to githeri cooking faster on a hot jiko. They also explain why both beakers end with the same amount of sulfur:	Mwalimu Achieng says: "Plot rate on the y-axis and temperature on the x-axis. Use the class means." She asks: "The demo at 50 C took 35 seconds. Where does that point sit on your graph? Is it where you expect?" and uses WAIT TIME of 10-15 seconds before cold-calling 3 learners. She then says: "Now go inside the beaker. Draw the particles at 10 C and at 55 C. Show how fast they move and show collisions." She circulates and prompts: "What does more kinetic energy mean for how often particles collide? And for how hard they hit?" She introduces the term: "The minimum energy a collision needs to react is the activation energy. A collision that has enough energy is an effective collision." She avoids saying that heat only makes collisions more frequent, and says: "Warm particles collide more often AND with more energy. The energy part matters most because a larger fraction of collisions are effective." She asks Amina and Kiprono to link the idea to the rusting-roof and uji ideas they posted on the DQB,	Modelling with particle diagrams and the Claim-Evidence-Reasoning (CER) structure. The claim is that higher temperature increases the rate; the evidence is the class table and graph, including how the 50 C demo value fits; the reasoning is the collision-theory model. Drawing the same beaker at two temperatures makes the invisible particle-level mechanism visible and testable against the data. Separating the speed of reaction from the amount of product avoids a common misconception.	Look for graphs with correct axes, labels and a rising curve. Check that learners place the 35 s demo ($28.6 \times 10^{-3} \text{ s}^{-1}$) between 19.2 and 38.5. In particle diagrams look for longer arrows at 55 C, more collisions, and at least one effective collision labelled. In written explanations check for all three ideas: more kinetic energy, more frequent collisions, and a larger fraction of effective collisions (activation energy). Flag any learner who claims warm particles are 'bigger', that the reaction makes more sulfur, or that temperature is a catalyst.

	temperature changes the speed, not the amount.	but keeps it brief and says those will be studied properly in Lesson 7. She asks: "Do the warm and cold beakers end up with the same amount of sulfur? Explain using particles." She cold-calls 4 learners and has partners rehearse their explanation aloud before writing it.		
Driving Question Board (DQB) Creation  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the DQB on the back wall. They review the questions from Lesson 1, including Wanjiru's first 'why' question, and decide with sticky notes which questions the temperature evidence now answers (for example, why warm Beaker B was faster). They write new questions raised by today's work, such as: Does cooling food, for example in a fridge, really slow reactions? Does doubling the amount of thiosulfate also change the time? Why does the rate roughly double for each 15 C rise? Would crushing a solid also help? They sort questions into 'answered by temperature', 'needs another factor' and 'needs more evidence', and add today's key evidence row (for example 25 C = 110 s; 55 C = 26 s) next to the answered ones.	Mwalimu Achieng says: "Wanjiru, read your first question again. Which part can we answer now, and which part can't we?" She gives WAIT TIME of 10-15 seconds, then invites 3 learners to move notes into the 'answered' column with a piece of evidence. She says: "Write at least one new question on a sticky note. Start with 'What would happen if...' or 'Why does...'" She cold-calls 4 learners to read their new questions aloud and group similar ones. She points out that questions on concentration and surface area will be investigated in the next lessons, and does not answer them now. She asks Amina and Kiprono whether their rusting-roof and uji questions have moved, and says that those will be revisited in Lesson 7.	Driving Question Board as a public record of growing understanding. Linking questions to evidence, and sorting them by what is answered and what is not, shows learners how investigations generate new questions and gives purpose to the next factor.	Check that new questions are testable and linked to variables (such as concentration or size of pieces) rather than just 'why is it hot?'. Note whether learners attach evidence (a time, rate or diagram) when moving a question to 'answered'. Count how many learners have posted at least one new question, and observe whether any question already points to concentration, surface area or catalysts.
Model Building Phase  VIDEO: Introduction to kinetics Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate	Learners open the model page in their science notebooks and revise the first model from Lesson 1. They draw the cold beaker and the warm beaker with particles, add arrows for speed, and label collisions, effective collisions and activation energy. Beneath the drawings they write a model statement: 'At a higher temperature particles have more kinetic energy, so they collide more often and more energetically, so a larger fraction of collisions are	Mwalimu Achieng says: "Your model from Lesson 1 had an idea about particles. Revise it today with kinetic energy, effective collisions and activation energy. Use a different colour pen for what you changed." She gives WAIT TIME of 10-15 seconds, then asks: "What is one thing your model can now explain that it couldn't before?" and cold-calls 3 learners. She asks 2 learners to read their model statement aloud while the class checks for the three ideas.	Cumulative model revision using an evidence-linked model statement and summary table. Revising the same model across lessons, with colour-coded changes, makes learning visible and shows that a model is improved by evidence. Tying each model feature to a data row keeps the explanation evidence-based and prepares learners to extend it to the next factor.	Check that revised models include speed arrows, collisions, an effective collision and the term activation energy. Check that the model statement has three parts (more kinetic energy, more frequent collisions, larger fraction effective) and that the evidence line quotes correct numbers (270 s and 26 s; 3.7 and 38.5 x10 ⁻³ s ⁻¹). Look for the note that the final amount of sulfur is unchanged. Use the summary table as an exit check: learners who cannot link

(Flexbook) Source: CK-12 Search ARES for similar readings	effective and the rate is higher.' They add the evidence line: 10 C = 270 s ($3.7 \times 10^{-3} \text{ s}^{-1}$) and 55 C = 26 s ($38.5 \times 10^{-3} \text{ s}^{-1}$), with the demo at 50 C = 35 s lying between 40 C and 55 C. They also add a note that the final amount of sulfur is the same, so temperature changes the speed only. They then complete the summary table (what we observed, what we learned, how we explain it) and add one question about how their model might also explain concentration.	She asks: "Where in your model does it show that both beakers end with the same amount of sulfur?" She then distributes the summary table and says: "Fill in all three columns with evidence from today's table." She collects notebooks for a quick check and ends by saying: "Next lesson we change something different, the concentration of the thiosulfate, and test whether the same collision idea still works."		observed times to the particle explanation need a short follow-up with the 10 C and 55 C diagrams.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners use the class data, not just intuition, to say that rate rises with temperature? Which pairs relied on 'heat makes things faster' without evidence? (2) Did anyone say that warm particles collide more often only? If so, plan a short follow-up on energy and the fraction of effective collisions. (3) Did anyone claim that a higher temperature gives more sulfur or that heat acts as a catalyst? Note names for Lesson 4 review. (4) Were safety routines (goggles, small quantities, ventilation, prompt rinsing, safe handling of warm baths) followed? (5) How well did learners place the demo's 35 s at 50 C relative to the 40 C and 55 C rows? (6) Which DQB questions now point naturally towards concentration for Lesson 4? Note which learners need support with $\text{rate} = 1/t$ calculations and graph plotting before Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Record the class-mean times for the cross to disappear at 10 C (270 s), 25 C (110 s), 40 C (52 s) and 55 C (26 s), and the rates 3.7 , 9.1 , 19.2 and $38.5 \times 10^{-3} \text{ s}^{-1}$. Note how the demo at 50 C (35 s) fits, and that the amount of sulfur made was the same.
What did I learn?	What did we learn? Write what happens to the rate of the thiosulfate-HCl reaction as temperature rises, and how the rate changes from 10 C to 55 C. State that temperature changes the speed of the reaction and not the final amount of product.
How does this explain the phenomenon?	How can we explain it? Use particles to explain why a higher temperature increases the rate: more kinetic energy, more frequent and more energetic collisions, a larger fraction of effective collisions (activation energy). Use this to explain why warm Beaker B was faster than cold Beaker A.

LESSON 4 (80 minutes (2 x 40-minute lessons)): Factor 2: Concentration

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how the concentration of sodium thiosulfate changes the rate of its reaction with hydrochloric acid, and use collision theory to explain why more particles per unit volume give more frequent collisions. This advances the Driving Question by showing a second factor that controls speed, without changing the temperature that explained Beaker B.
Knowledge	Learners explain that a higher concentration means more particles per unit volume, so collisions are more frequent and more effective collisions happen each second. They state that rate is roughly proportional to concentration, using the data: $0.15 \text{ mol/dm}^3 = 110 \text{ s}$ ($9.1 \times 10^{-3} \text{ s}^{-1}$); $0.12 = 138 \text{ s}$ (7.2); $0.09 = 183 \text{ s}$ (5.5); $0.06 = 275 \text{ s}$ (3.6). They state that concentration changes the speed, not the final amount of sulfur.
Skills	Planning a fair test by naming independent, dependent and control variables; measuring time to the cross disappearing; converting time to rate ($\text{rate} = 1/t$); plotting rate (y-axis) against concentration (x-axis); describing a trend and checking proportionality ($\text{rate} / \text{concentration}$ is about 60); using a particle diagram as a model (SEP: Developing and Using Models; Planning and Carrying Out Investigations; Analyzing and Interpreting Data).
Attitudes	Respect: learners listen to and value the ideas of group members when discussing patterns. Responsibility: learners take care with acid, goggles and ventilation. Curiosity about why a stronger solution reacts faster, and appreciation of how controlling concentration is useful in daily life (for example antacid tablets and cleaning products). Core competencies: Communication and collaboration; Critical thinking and problem solving. PCI: Safety and security.
Key Inquiry Question	KICD Key Inquiry Question: How do various factors affect the rate of reactions?
Purpose in Storyline	Lesson 3 gave the first statement of collision theory (higher temperature means more energetic and more frequent collisions, so more are effective) and the 25 C result of 110 s. Lesson 2 gave $\text{rate} = 1/t$ and the variables of a fair test. Lesson 4 keeps temperature at 25 C and changes a different factor, concentration, so learners see that the same collision idea (effective collisions per second) explains a second factor. It prepares Lesson 5 (surface area).
Safety Notes	Wear goggles. Sulfur dioxide (SO_2) is an irritant: work in a well-ventilated room, use only the small volumes stated (10 cm ³ of 1.0 mol/dm ³ HCl, up to 50 cm ³ of thiosulfate solution), and rinse each beaker in the sink straight after the cross vanishes. Wipe spills at once; wash hands after the practical. Students with breathing problems such as asthma should observe from a distance and record data. Do not smell the mixtures. Keep all tests at 25 C and use the room-temperature water bath.

B. LESSON OVERVIEW

Mwalimu Achieng reopens the Driving Question Board (DQB) and recalls the Disappearing Cross: Beaker B (50 C) lost its cross at 35 s and Beaker A (10 C) at 270 s, with the same chemicals in the same amounts. Lesson 3 showed that temperature changes the rate. Learners now ask whether anything else besides temperature could change the time. Using the antacid tablet fizzing in a glass of water (a whole tablet in a small amount of water compared with a dilute solution) as a hook, they predict what happens to the cross time when the thiosulfate solution is made weaker, using water. In groups they run the cross test at 25 C with four concentrations (0.15, 0.12, 0.09, 0.06 mol/dm³), keeping 10 cm³ of 1.0 mol/dm³ HCl, the total volume of 50 cm³, the beaker and the observer constant. Class-mean results are 110 s, 138 s, 183 s and 275 s, giving rates 9.1, 7.2, 5.5 and $3.6 \times 10^{-3} \text{ s}^{-1}$. Learners plot rate against concentration, notice the straight-line trend through the origin, check that $\text{rate} / \text{concentration}$ is about 60 for every row, and explain with particle diagrams: more particles per unit volume, so more frequent collisions and more effective collisions per second. They also note that the final amount of sulfur depends on the amount of thiosulfate, not on the speed. The DQB is updated and the class model gets a second

factor.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Learners recall Beaker A and B (35 s vs 270 s). Mwalimu Achieng shows an antacid tablet fizzing in a glass: she breaks one tablet and drops it in 100 cm ³ of water. She then asks: if we kept the temperature the same but made the sodium thiosulfate solution weaker by adding water, what would happen to the time for the cross to disappear? Individually, learners write in their science notebooks: (1) a prediction (faster, slower or the same), (2) a reason with a particle sketch, and (3) a rough guess of the time for 0.06 mol/dm ³ compared with 110 s for 0.15 mol/dm ³ . Wanjiru shares first. Learners then pair-share, and Otieno links the answer to his earlier particle idea.	Say: 'Last lesson, temperature explained why Beaker B beat Beaker A. Today the temperature stays at 25 C. What else could change the time?' Show the fizzing antacid and ask: 'Think of the fizz. What would you change to make it fizz slower or faster, other than heating?' Give 1 minute silent writing, then WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 learners (including Wanjiru and Otieno) for predictions and record the three kinds of prediction (faster/slower/same) as tally marks on the board. Ask: 'Draw the particles in a strong solution and in a weak solution. What differs?' Do NOT confirm any answer. Say: 'We will test it, and we will keep the cross in view.'	Predict-Observe-Explain with particle sketches: writing a prediction and a reason before data exposes learners' existing models, so later evidence confirms or challenges something they personally committed to. Tally marks make the class's range of ideas visible without judging.	Look for: every learner has a written prediction, a reason and a particle sketch. Indicators: sketches show different numbers of particles in the same volume (strong sign of concentration idea); predictions that mention 'more particles' or 'more collisions'; misconceptions such as 'weaker solution means less sulfur made faster' or 'concentration changes temperature'. Record tally counts of faster/slower/same. Note learners who link back to Lesson 3 collision ideas.
Observe Phase  VIDEO: Factors impacting rates of reactions - Overview Source: CK-12  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4 (roles: reader, timer, pourer, recorder), learners plan and run the cross test at 25 C. First they fill a planning table in their notebooks: Independent variable = concentration of sodium thiosulfate (0.15, 0.12, 0.09, 0.06 mol/dm ³); dependent variable = time for the cross to disappear (converted to rate = 1/t); control variables = 10 cm ³ of 1.0 mol/dm ³ HCl, 50 cm ³ total volume of solution, temperature 25 C, same beaker type, same black cross, same observer, same bench lighting. They prepare the thiosulfate by mixing volumes: 50 cm ³ thiosulfate + 0 water; 40 + 10;	Hand out goggles and check every pair before any acid is poured: 'Goggles on, hair tied back, windows open.' Remind: 'SO ₂ is an irritant. Small amounts only, and rinse the beaker the moment the cross is gone.' Circulate and ask each group: 'Which variable are you changing? Which are you keeping the same? How do you know the cross is gone?' (Cross is gone when the observer looking down can no longer see any part of the black lines.) Check that the same person judges the cross in every run. After runs, collect times on the board and compute the class mean. Ask: 'Which	Structured data collection with a shared class table: pooling results reduces the effect of individual error, shows reliability (the 0.15 row matches Lesson 3), and gives every learner the same evidence for sensemaking. Naming variables before the practical makes the fair test explicit (SEP: Planning and Carrying Out Investigations).	Observe: correct dilution volumes in each beaker; goggles worn; same observer and same lighting; timer stopped at the same cross-visibility criterion; beaker rinsed immediately. Check notebooks for correct identification of independent, dependent and control variables, and for correct 1/t calculations (9.1, 7.2, 5.5, 3.6 x10 ⁻³ s ⁻¹). Ask a quick thumbs check: 'Did the weaker solution take longer?' Note groups whose times are far from the class mean and ask them to suggest why (measurement error, uneven mixing, different observer).

	30 + 20; 20 + 30. They put the cross under the beaker, add 10 cm³ HCl, start the stopwatch, look down from above and stop when no part of the black lines can be seen, then rinse the beaker straight away. Groups pool their times with the class. Class-mean values: 0.15 mol/dm³ = 110 s; 0.12 = 138 s; 0.09 = 183 s; 0.06 = 275 s. They calculate rate = 1/t in x10⁻³ s⁻¹: 9.1, 7.2, 5.5 and 3.6. The 0.15 result matches the 25 C row from Lesson 3 (110 s). Learners record the table.	concentration took the longest? Does the 0.15 row match something we measured in Lesson 3?' Wait 10-15 seconds, then cold-call 4 groups to report their time for one concentration. Ask learners to convert to rates and write them in the table with units x10⁻³ s⁻¹. Say: 'Rate here is a relative rate: 1 divided by the time.'		
Explain Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Factors Affecting Reaction Rate Practice Source: CK-12  Search ARES for similar readings	Learners plot the class data on graph paper: concentration (mol/dm³) on the x-axis and rate (x10⁻³ s⁻¹) on the y-axis, as in Lesson 2. They draw a best-fit line and describe the trend: as concentration decreases from 0.15 to 0.06, rate falls from 9.1 to 3.6, and the points lie close to a straight line through the origin. They calculate rate / concentration for each row (about 60 each time) and conclude that rate is roughly proportional to concentration: halve the concentration and the rate roughly halves. Then they explain with particle diagrams: the same volume (50 cm³) contains more thiosulfate particles at 0.15 than at 0.06, so thiosulfate and H⁺ particles meet more often, giving more collisions per second and so more effective collisions per second. Because the temperature is the same, the energy of each particle is the same, so the fraction of collisions that are effective does not change. Learners also compare the two ideas: temperature changed both the energy and the number of collisions (Lesson 3), but	Say: 'Plot concentration on the x-axis and rate on the y-axis, as we did in Lesson 2. What does the pattern tell you?' Ask: 'Describe the line in one sentence using numbers from the table.' Wait 10-15 seconds. Cold-call 3 learners. Ask: 'What happens to rate / concentration in each row? Calculate 9.1 / 0.15 and 3.6 / 0.06.' Guide to about 60 and say: 'This means rate is roughly proportional to concentration.' Then ask: 'Look at your particle diagrams from the Predict Phase. What does 0.15 have more of in 50 cm³?' (More particles per unit volume.) Press for the chain of reasoning: 'More particles per unit volume, so what happens to collisions? Then what happens to effective collisions per second?' Cold-call 3 more learners to complete the sentence frame: 'Because there are more ____ in the same volume, ____ happen more often, so ____.' Ask: 'Did the particles gain energy as we diluted or concentrated the solution? How do you know?' (Temperature was constant at 25 C.) Ask: 'In Lesson 3 energy changed. What is different here?' Correct any claim	Claim-Evidence-Reasoning with particle models: learners state the claim (rate roughly proportional to concentration), cite the data (rate / concentration is about 60), and give the collision reasoning (more particles per unit volume, more frequent collisions, more effective collisions per second). Comparing with Lesson 3 separates frequency of collisions (concentration) from energy of collisions (temperature), which prevents the misconception that frequency means energy (SEP: Analyzing and Interpreting Data; Constructing Explanations).	Check graphs: axes labelled with units, correct plotting, best-fit line through or close to the origin. Check statements: 'rate is roughly proportional to concentration' backed with the ratio of about 60 for every row. Check explanations for the three-step chain (more particles per unit volume, so more frequent collisions, so more effective collisions per second) and no confusion with energy. Ask a mini-whiteboard question: 'If the concentration doubled from 0.06 to 0.12 mol/dm³, what happens to the rate? (3.6 to 7.2 x10⁻³ s⁻¹, roughly double).' Indicators of mastery: 80% of learners write the correct chain; misconceptions noted for follow-up.

	concentration changes only how many collisions occur per second. They link back to the antacid: a stronger acid solution fizzes faster. They finally decide whether the amount of sulfur at the end would differ (the final amount depends on the amount of thiosulfate: a weaker solution contains less thiosulfate but the speed is not the same as the amount).	that a higher concentration means particles move faster. Ask Kiprono to say how a stronger cup of tea or uji mix tastes sooner, then link carefully: 'Is that the same idea? What changed, the number of particles in the volume or their energy?' Close with: 'Would both beakers still end with sulfur?' (Yes: concentration changes the speed, not the type of product.)		
Driving Question Board (DQB) Creation  VIDEO: Factors impacting rates of reactions - Overview Source: CK-12  Search ARES for similar videos  HTML: Molarity Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners return to the DQB on the back wall. They add sticky notes with new questions arising from today's work, for example: 'Does doubling concentration always double the rate?', 'Why was the 0.06 mol/dm ³ time (275 s) almost the same as Beaker A at 10 C (270 s)?', 'Can a gas be made more concentrated by pressing it?', 'Why does a crushed antacid tablet fizz faster than a whole one?' Groups sort questions into 'answered today' (for example, why was the weaker solution slower), 'still to investigate' (surface area, catalysts, light) and 'new'. Learners move a question or two that were answered today into the 'Answered' column with a one-line evidence note citing the table. They keep their own copy in their notebook.	Say: 'Write one question that today's data raised for you, on a sticky note.' Allow 2 minutes of silent writing and then WAIT TIME of 10-15 seconds before sharing. Cold-call 4 learners to read a question and place it on the DQB. Ask: 'Which of our earlier questions about Beaker A and B can we now partly answer? What is the evidence?' Invite Wanjiru to revisit her first 'why' question and say what part is explained so far. Ask Amina and Kiprono whether their rusting roof and tea/uji examples relate to concentration: 'Would a more concentrated solution change how fast things happen? Hold that thought for Lesson 7.' Point out the 'crushed tablet' question: 'That one leads us straight to Lesson 5.' Accept all questions respectfully and do not answer them now.	Driving Question Board as a public record of growing understanding: moving questions between 'still to investigate' and 'answered' with evidence makes progress visible, and new questions create a purpose for the next lesson. Learners see science as a cycle of questions and evidence.	Count and review sticky notes: at least one question per learner. Indicators: questions that use particle language (particles, collisions, volume), questions that propose a new factor (surface area, pressure), and questions that connect to Kenyan contexts. Check that learners cite table evidence when they move a question to 'Answered'. Note learners writing only factual or off-topic questions for support.
Model Building Phase  VIDEO: Introduction to reaction rates <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Learners revise their class collision model started in earlier lessons. On a poster or in their notebooks they draw two labelled boxes of equal volume at 25 C: a high-concentration box (0.15 mol/dm ³) with many thiosulfate and H ⁺ particles and a low-concentration box (0.06 mol/dm ³) with few. They add arrows showing collisions and shade the effective ones. They	Say: 'Add today's factor to our model. Show it with particles in equal volumes.' Circulate and ask: 'Where are the effective collisions? Where did energy change, and where did only the number of particles change?' Ask: 'Predict the rate for 0.075 mol/dm ³ . What evidence supports your prediction?' WAIT TIME 10-15 seconds, then cold-call 3 learners.	Cumulative model revision with prediction: adding concentration to the collision model, and then using it to predict an unseen value (0.075 mol/dm ³) tests whether the model is useful, not just descriptive. Explicitly returning to the anchor demo reinforces that the model must explain the phenomenon (SEP: Developing and Using Models).	Check models for: equal-volume boxes with different numbers of particles, collision arrows, effective collisions marked, and a clear note that energy was unchanged. Check predictions: about half the rate at about half the concentration (around 4.5 x10 ⁻³ s ⁻¹) with the reasoning from the table. Exit question: 'Does a weaker solution make less sulfur at the end, or just

Search ARES for similar videos HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12 Search ARES for similar readings	add a line to the model: 'Higher concentration = more particles per unit volume = more frequent collisions = more effective collisions per second = higher rate (roughly proportional).' Under the temperature factor from Lesson 3 they add: 'Concentration changes how often particles meet; it does not change their energy.' They then use the model to answer: (a) Explain why the 0.06 mol/dm³ solution at 25 C took 275 s while 0.15 took 110 s. (b) Predict the rate if the concentration were 0.075 mol/dm³ (about half of 0.15, so roughly half the rate, about 4.5 x10⁻³ s⁻¹). (c) Link to the antacid tablet fizzing in a small volume of water versus a large volume. They complete the ARES Summary Table and write one sentence linking the model to the anchor demo (Beaker A and B), explaining that concentration was the same in both beakers, so temperature was the cause.	Then ask: 'In the anchor demo both beakers had the same concentration. Which factor explains the difference? Which factor is ruled out?' Have learners complete the Summary Table. Ask one pair to explain the model to another pair. Close: 'Next lesson: why does a crushed antacid tablet fizz faster than a whole one?' Collect notebooks or photograph models for the teacher reflection.		make it more slowly?' Look for the idea that concentration changes the speed (more precisely, the amount depends on how much thiosulfate is present). Indicators of mastery: model uses the term 'effective collisions per second' correctly and does not use 'frequency' to mean 'energy'.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners separate the effect of concentration (number of collisions per second) from temperature (energy and number of collisions)? Which learners still say that 'stronger' means 'particles move faster'? (2) Did every group control the variables (10 cm³ HCl, 50 cm³ total, 25 C, same observer and lighting)? Which results were far from the class mean, and why? (3) Could learners explain why rate / concentration is about 60 for every row and use it to say 'roughly proportional'? (4) Did learners use the three-step chain (more particles per unit volume, more frequent collisions, more effective collisions per second)? (5) Were safety routines followed: goggles, ventilation, small volumes, immediate rinsing? (6) Which DQB questions show readiness for surface area in Lesson 5? What would you change in time management, grouping or the antacid hook next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that at 25 C the cross disappeared in 110 s, 138 s, 183 s and 275 s as the sodium thiosulfate concentration decreased from 0.15 to 0.12, 0.09 and 0.06 mol/dm ³ . The matching rates (1/t) were 9.1, 7.2, 5.5 and 3.6 x10 ⁻³ s ⁻¹ . The weaker the solution, the longer the cross stayed visible.
What did I learn?	We learned that a higher concentration means more particles per unit volume, and that rate is roughly proportional to concentration

	(rate / concentration is about 60 for every row). Concentration changes how fast the reaction goes, not the type of products, and it is a second factor besides temperature.
How does this explain the phenomenon?	We explained that in a more concentrated solution there are more particles in the same volume, so they collide more often, giving more effective collisions per second and a faster rate. The energy of the particles did not change because the temperature stayed at 25 C. In the anchor demo the concentration was the same in Beakers A and B, so temperature, not concentration, caused the 35 s versus 270 s difference.

LESSON 5 (80 minutes (2 x 40-minute lessons)): Factor 3: Surface Area – Why Does Powder React Faster Than Chunks?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use gas-volume data to show that breaking a solid into smaller pieces increases the rate of reaction, explain this using collision theory, and read a rate from the gradient of a volume-time curve. This extends the anchor question: if temperature and concentration change how often and how energetically particles collide, what else can change the number of effective collisions per second?
Knowledge	By the end of the lesson, the learner should be able to: (a) describe rate of chemical reactions as the change in amount of reactant or product per unit time, measured here as volume of CO ₂ (cm ³) per second; (b) explain that smaller pieces expose more surface, so more particles of the solid are available for collisions with acid particles and there are more effective collisions per second; (c) state that the gradient of a volume-time curve is the rate; (d) state that surface area changes the speed of the reaction, not the final amount of gas made (maximum about 120 cm ³).
Skills	Perform experiments involving the rate of chemical reactions (set up gas collection, take readings every 30 s); identify and control variables (mass 0.5 g, 25 cm ³ of 1.0 mol/dm ³ HCl, 25 °C); record and tabulate data; plot volume-time graphs; calculate initial rate = volume/time; draw and evaluate conclusions from evidence (NGSS practices: Planning and Carrying Out Investigations, Analyzing and Interpreting Data, Constructing Explanations, Developing and Using Models).
Attitudes	Appreciate the importance of manipulating factors affecting the rate of chemical processes in day-to-day life (for example unga versus whole maize for uji, crushed antacid tablets). Core competencies: Communication and collaboration (discussing factors with peers) and Critical thinking and problem solving. Values: Responsibility and Respect for others' opinions. PCI: Safety and security when carrying out activities to investigate rates of reactions.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lessons 3 and 4 gave us collision theory: higher temperature and higher concentration produce more effective collisions per second. Lesson 1 introduced gas volume as one of three ways to measure rate. Here learners test a third factor with a new measuring method, apply the same collision idea to a solid, and learn to read rate as a gradient. This prepares Lesson 6, where learners meet a rate change that does not come from a change in temperature, concentration or particle size.
Safety Notes	Wear goggles throughout. 1.0 mol/dm ³ HCl is an irritant: handle with care, wipe spills at once and rinse splashes on skin with plenty of water. Do not put powdered chalk or marble dust near eyes or breathe it in. Use a bung or delivery tube fitted loosely so pressure cannot build up. Keep the room ventilated. Report any spills or injury to Mwalimu Achieng and use the first aid kit. Wash hands after the practical and dispose of used mixtures as instructed.

B. LESSON OVERVIEW

Mwalimu Achieng reopens the Driving Question Board (DQB) and reminds the class that the cross vanished at 35 s in warm Beaker B and at 270 s in cold Beaker A, with the same chemicals in the same amounts. Temperature (Lesson 3) and concentration (Lesson 4) both changed the number of effective collisions per second. Learners now ask: can the SIZE of a solid change the rate even when temperature, concentration and amounts are the same? Using the Kenyan example of unga versus whole maize for uji and a fizzing antacid tablet, learners predict, then test 0.5 g of marble chips (CaCO₃) as large chips (about 10 mm), small chips (about 3 mm) and powder in 25 cm³ of 1.0 mol/dm³ HCl at 25 °C, collecting CO₂ over 120 s. Class-mean data show that powder is fastest and levels off at 120 cm³. Learners calculate initial rates, read gradients on volume-time curves, explain with collision theory, post new DQB questions and revise their cumulative model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Learners recall the anchor demo (Beaker B 35 s, Beaker A 270 s) and the Lessons 3 and 4 explanation of effective collisions. Mwalimu Achieng holds up a whole antacid tablet and a crushed one, and shows a handful of whole maize kernels next to unga. Learners individually write in their science notebooks: (1) Which will react faster with acid, a big marble chip or the same mass of marble powder? (2) Draw the CO ₂ volume-time line you expect for large chips, small chips and powder on one sketch graph. (3) Do you think they will make the same TOTAL amount of gas? Why? Then they share with a partner and agree on one prediction. Kiprono from Kericho wonders whether unga makes uji ferment faster than whole maize, and Amina asks if this has anything to do with the cross in the beakers.	Say: "Last lesson we saw that more particles in the same volume means more collisions. Today the acid and the chemicals stay the same. I am only changing one thing: how the marble is cut up. Before I show you anything, commit to a prediction." Hold up the crushed and whole tablet. Give WAIT TIME of 10-15 seconds of silent writing. Then say: "Turn to your partner and compare sketch graphs. You have two minutes." Cold-call 4 pairs: "Which sample did you put highest, and what made you choose it?" Record predictions on the board in three columns (large, small, powder) without saying who is right. Ask: "What do you think is the same in all three tests?" and let learners list the control variables. Do not explain surface area yet.	Predict-Commit-Compare: committing to a written prediction and a sketch graph before the evidence makes learners' preconceptions visible, so the later data can confirm or challenge them. Asking about the total amount of gas surfaces the misconception that a faster reaction makes more product (linked to the anchor outcome that both beakers end with the same amount of sulfur).	Check notebooks for: (a) a sketch graph with a prediction for all three samples; (b) a reason that mentions pieces, size or 'more of it touching the acid'; (c) a statement about whether the total gas is the same. Note learners who predict a bigger final volume for powder. Listen for the language of Lessons 3-4 (collisions, particles). Tally on a clipboard how many predict powder fastest.
Observe Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4 with assigned roles (reader, timer, gas-syringe or measuring-cylinder operator, recorder), learners wear goggles and carry out the practical at room temperature (about 25 C). For each test they add 0.5 g of marble chips (CaCO ₃) to 25 cm ³ of 1.0 mol/dm ³ HCl in a conical flask, quickly fit the bung and delivery tube, and record the volume of CO ₂ collected (cm ³) at 30 s, 60 s, 90 s and 120 s. The equation is CaCO ₃ (s) + 2HCl(aq) → CaCl ₂ (aq) + H ₂ O(l) + CO ₂ (g). Groups test large chips (about 10 mm), small chips (about 3 mm)	Before starting say: "Name the independent variable, the dependent variable and three control variables. Thumbs up when you have all five." Cold-call 3 groups. Circulate and check goggles, bungs and that the timer starts the moment the marble touches the acid. Ask each group: "What is your timer's job at 30 seconds?" and "How do you know the acid is the same in each test?" Remind: "Add the powder quickly and keep it away from your face." Give a 30-second scribe count so that all groups read together. Pause after the data are on the	Structured Data Collection with Noticing and Wondering: roles keep all learners active, and a silent noticing routine focuses attention on patterns in the table (ordering of rows, the plateau at 120 cm ³) before any explanation is offered. Control-variable naming rehearses the fair-test thinking from Lesson 2.	Observe whether groups: wear goggles and fit bungs safely; keep mass (0.5 g), acid (25 cm ³ of 1.0 mol/dm ³) and temperature the same; read volumes at the right times; and record in a neat table. In the noticing round listen for 'powder fizzed most at the start', 'powder stopped at 120' and 'large chips are slowest'. Note groups whose results differ and ask why (timing error, spillage, sample variation).

	and powder. They watch the bubbling closely and note which sample fizzes most at the start. Class-mean results are entered on the board table: Large chips: 12, 22, 31, 39 cm³. Small chips: 28, 50, 68, 82 cm³. Powder: 62, 100, 117, 120 cm³. Powder levels off at about 120 cm³, the maximum possible from 0.5 g (0.005 mol) because HCl is in excess (0.025 mol). Learners note in their science notebooks that the mass of marble, the volume and concentration of acid and the temperature stayed the same.	board and say: "Look at the table in silence for 15 seconds. Write one thing you notice and one thing that surprises you." WAIT TIME 10-15 seconds, then cold-call 3 learners. Ask: "Which row levelled off? What number did it stop at?" Do NOT explain yet. Rinse flasks and clear the bench; collect used materials in the waste container.		
Explain Phase  VIDEO: Collision theory <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Factors Affecting Reaction Rate Practice Source: CK-12  Search ARES for similar readings	Learners first plot volume of CO₂ (y-axis) against time (x-axis) for all three samples on one set of axes using the class-mean data, joining points with smooth curves. They compare the sketch they predicted with the real graph. Next they calculate the initial rate over the first 30 s: large 12/30 = 0.40 cm³/s; small 28/30 = 0.93 cm³/s; powder 62/30 = 2.07 cm³/s. They draw a tangent or straight line along the first part of each curve and see that the steeper the line, the greater the rate: the gradient (slope) of a volume-time curve is the rate. They explain why powder levels off at 120 cm³ while the others are still rising at 120 s (the marble in powder is used up first; all three would give about 120 cm³ eventually). Using particle diagrams, learners draw a large chip and the same mass of powder, shading the particles on the outside that can meet acid particles, and write a collision-theory explanation: smaller pieces expose more surface, so more CaCO₃ particles are touched by HCl particles each second, giving	Say: "Plot volume on the y-axis and time on the x-axis. Use a different symbol for each sample." Circulate and ask: "Which curve is steepest at the start? What does steep mean for rate?" Then say: "The gradient is the rate. Show me with your hand how the steep part and the flat part of the powder curve differ." WAIT TIME 10-15 seconds before taking answers. Cold-call 3 learners to calculate each initial rate at the board: 0.40, 0.93, 2.07 cm³/s. Ask: "Why does the powder curve flatten at 120 cm³?" and cold-call 2 learners; guide: "What is limiting the gas, the acid or the marble?" (HCl is in excess, 0.025 mol versus 0.005 mol of CaCO₃). Then ask: "Use the words particles, surface, collisions and effective in one sentence." Cold-call 4 learners and have the class refine one model sentence. Challenge: "Is the powder particle itself any hotter or more energetic? Where did the extra collisions come from?" Reinforce that surface area changes the number of collisions per second, not the energy per	Claim-Evidence-Reasoning with Model-Based Explanation: learners make a claim (smaller pieces react faster), cite evidence (initial rates 0.40, 0.93, 2.07 cm³/s; steeper curves; powder reaches 120 cm³ first), and reason with collision theory (more exposed particles, more effective collisions per second). Reading the gradient as the rate links the graph to the rate definition and connects the gas method from Lesson 1 to the cross method (rate = 1/t).	Check graphs for correct axes (volume on y, time on x), labelled curves and a plateau at 120 cm³. Check calculations: 12/30 = 0.40, 28/30 = 0.93, 62/30 = 2.07 cm³/s with units. Listen for 'steeper gradient = faster rate' and for explanations that use effective collisions per second without claiming the particles have more energy. Ask three learners to explain the plateau; look for the idea that the marble is used up and the acid is in excess. Check particle diagrams show more exposed surface for the powder.

	more effective collisions per second. They compare it with Lessons 3 and 4 and note that here temperature and concentration did not change, so the particles' energy and the acid concentration are the same; only the number of particles available to collide changed. They link back to the cross: a larger exposed surface would let the thiosulfate cross vanish sooner only if a solid were involved, but there the factor was energy. They also apply it to uji: unga has far more surface than whole maize, so it cooks and ferments faster, and Kiprono's question gets a first answer.	collision. Correct the misconception that more surface makes more total gas.		
Driving Question Board (DQB) Creation  VIDEO: Collision theory and the Maxwell-Boltzmann distribution <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the DQB on the back wall, kept by Mwalimu Achieng. Wanjiru reads the original question again (why was Beaker B faster?). Groups look for questions already answered and move them to an 'Answered' area with a note of the evidence (for example 'Particle size changes rate: powder 2.07 cm³/s vs large chips 0.40 cm³/s'). Each group writes at least one new question on a sticky note. Possible questions: Why do antacid tablets work faster when crushed or chewed? Does surface area matter for maize flour dust catching fire in a mill? Why does unga cook faster than whole maize? Is there a factor that speeds up a reaction without changing temperature, concentration or size? Do the factors work together? Learners place their notes beside the matching factor heading (temperature, concentration, surface area, catalyst, light).	Say: "Look at our DQB. Which questions can we now answer with evidence from today? Come up and move one, and tell us the data you used." Cold-call 3 learners to move a question. Then say: "Write one new question that today's evidence made you wonder about, in the form 'What would happen if...' or 'Why does...'. Take 2 minutes." WAIT TIME 10-15 seconds before sharing. Cold-call 4 groups to read their questions aloud. Group questions by theme and ask: "What do these questions have in common?" Point out the question about a speed-up with no change to temperature, concentration or size and say: "Hold on to that one for next lesson." Thank learners for respectful listening to others' opinions.	Driving Question Board as a living record: moving answered questions with cited evidence builds the habit of evidence-based answers, while new questions keep the inquiry going and point to the next investigation (Lesson 6: catalysts and light).	Look for each group moving at least one question with a correct evidence statement from today's table or graph. Check new questions are testable and connected to the phenomenon, not simple recall. Note learners who ask about a factor not yet covered, as a sign of readiness for Lesson 6.
Model	Learners open the cumulative	Say: "Open your model page. Add	Cumulative Model Revision:	Check the model panel for: more

Building Phase  VIDEO: Surface area to volume ratio of cells <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12  Search ARES for similar readings	model page in their science notebooks, started in Lesson 1 and revised in Lessons 3 and 4. They add a third panel: 'Surface area'. Using particle drawings, they show a large chip with few exposed particles and powder with many exposed particles, with acid particles colliding at the surface. They label 'more surface = more effective collisions per second'. In a summary line they write that temperature changes the energy and frequency of collisions, concentration changes how crowded the particles are, and surface area changes how many solid particles can be reached. They add the evidence from the table: initial rates 0.40, 0.93 and 2.07 cm³/s, and the gradient idea. They revisit the anchor demo and write one sentence on why the warm Beaker B (35 s) beat cold Beaker A (270 s), 7.7 times faster, with the same chemicals and same final amount of sulfur. They also note one limit of the model: surface area did not change the final volume (120 cm³). Finally they write a short real-life application: why unga cooks faster than whole maize or why a crushed antacid tablet acts quickly, and one safety idea such as keeping flour dust away from flames. Learners share their models with a partner and give one piece of kind feedback.	a new panel for surface area. Draw the chip, the powder and the acid particles, and label where collisions happen." Circulate with the question: "Where on this drawing can an acid particle actually hit a marble particle?" WAIT TIME 10-15 seconds before pairs share. Cold-call 3 learners to explain their panel using the words particles, surface and effective collisions. Then say: "Now connect all three factors so far. In one sentence, what do they all change?" Guide to 'the number of effective collisions per second'. Ask: "What does surface area NOT change?" (total amount of product). Collect 3 exit tickets: 'Powder fastest because...' and 'The gradient of a volume-time curve is...'. Preview Lesson 6: "Next lesson we will see a reaction speed up with no change to temperature, concentration or size."	learners extend the same particle model across factors, which shows the unifying idea of effective collisions per second, and they test the model against the anchor demo. Stating the limits of the model (final amount unchanged) strengthens scientific reasoning and prepares for the catalyst lesson.	exposed particles on powder, labelled collisions at the surface, the phrase effective collisions per second, and evidence figures (0.40, 0.93, 2.07 cm³/s). Check the exit ticket: powder is fastest because it has the largest surface area, and the gradient equals the rate. Note learners who still say a bigger surface gives more product; follow up with the 120 cm³ plateau. Check the anchor-demo sentence matches the facts (35 s vs 270 s, 7.7 times).
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D. TEACHER REFLECTION

Which learners explained the rate difference with surface and collisions rather than saying 'powder is smaller so it is quicker'? Did groups control the variables (0.5 g, 25 cm³ of 1.0 mol/dm³ HCl, 25 °C) and time the first 30 s carefully, since initial rate depends on it? Did anyone confuse the steeper gradient with a larger final volume, or claim the powder particles have more energy? Which graphs had wrong axes or no plateau? Did the DQB questions hint at readiness for catalysts? What would I change: more practice reading gradients, a shorter practical so more time for discussion, or an extra Kenyan example such as flour dust or crushed antacid? Record

learners who need support with plotting and calculations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Powder fizzed fastest and levelled off at 120 cm ³ of CO ₂ ; small chips were next; large chips were slowest. Initial rates over the first 30 s were 0.40 cm ³ /s (large), 0.93 cm ³ /s (small) and 2.07 cm ³ /s (powder).
What did I learn?	Breaking a solid into smaller pieces increases its surface area, so more particles can collide with acid particles and the rate rises. The gradient of a volume-time curve gives the rate. Surface area changes the speed, not the final amount of gas (maximum about 120 cm ³).
How does this explain the phenomenon?	With the same mass, acid and temperature, powder exposes more CaCO ₃ particles to HCl particles, so there are more effective collisions per second and a steeper curve. This is the same collision idea used for temperature and concentration, and it helps explain why unga cooks faster than whole maize and why a crushed antacid works quickly.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Factor 4: Catalysts and Light

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how a catalyst changes the rate of a reaction without a change in temperature, concentration or size of pieces, and use collision theory to explain why. They also recognise that light supplies energy for some reactions. This answers the phenomenon question of what controls speed inside the beaker and extends it to the enzymes in the human gut and to industry.
Knowledge	By the end of the lesson the learner should be able to: (a) describe rate of chemical reactions; (c) explain factors that influence the rate of reactions in biological, chemical and physical processes (catalyst and light). Specifically: a catalyst, such as the enzyme catalase in liver, speeds up a reaction by providing an alternative pathway with lower activation energy (the minimum energy needed for a collision to react) and is not used up. An enzyme is a biological catalyst. Light is a source of energy for some reactions (photosynthesis, silver bromide in photographic film) and is not a catalyst. Factors change the speed of a reaction, not the final amount made.
Skills	NGSS SEP: Planning and Carrying Out Investigations (identify independent, dependent and control variables in the catalyst demo); Analysing and Interpreting Data (compare foam heights of 0, 0, 0.5 and 9 cm); Constructing Explanations (use collision theory and activation energy); Developing and Using Models (add the catalyst to the particle model). Also observing, measuring (foam height in cm), communicating and collaborating.
Attitudes	Values (KICD): Responsibility and Respect: carefully uses materials and respects others' opinions during group discussion of factors affecting the rate of reactions. Appreciates the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life. Core competencies: Communication and collaboration; Digital literacy (research on optimum conditions in biological, chemical and physical processes). PCI: Safety and security: observes safety measures when carrying out activities to investigate rates of reactions.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	In Lessons 3 to 5 the only factors were temperature, concentration and surface area, each changing the frequency or energy of collisions. Here learners meet a rate change with no change in temperature, concentration or size, which challenges the model and pushes it forward: a catalyst lowers the activation energy so a larger fraction of collisions are effective. This prepares Lesson 7 (optimum conditions for enzymes) and Lesson 8 (final model).
Safety Notes	Wear goggles. 3% hydrogen peroxide can irritate eyes and skin; rinse any splash with plenty of water. Teacher handles the peroxide and the liver. Wash hands after touching raw liver and keep it away from the mouth. Use a tray under the measuring cylinders for foam overflow. Boil the liver cube in advance with teacher supervision only. Dispose of the liver and solutions as the teacher directs. No tasting of any material. For the discussion, note that SO ₂ from the anchor reaction is an irritant, so that reaction is not repeated here.

B. LESSON OVERVIEW

Mwalimu Achieng returns to the Disappearing Cross: warm Beaker B (50 C) was 7.7 times faster than cold Beaker A (10 C), 35 s versus 270 s, with identical chemicals. Lessons 3 to 5 gave temperature, concentration and surface area, all of which change how often or how energetically particles collide. Today Wanjiru, Otieno, Amina and Kiprono's groups meet a puzzle: 10 cm³ of 3% hydrogen peroxide with a few drops of detergent shows no foam on its own, yet a 1 cm³ cube of fresh liver makes a column of foam 9 cm high within 30 s at the same temperature, concentration and amount. Learners predict, observe the four set-ups (no additive, sand, boiled liver, fresh liver), find that the liver is not used up, and explain using catalyst, enzyme (catalase) and lower activation energy via an alternative pathway. They then discuss the Haber process (iron catalyst for ammonia, used worldwide; Kenya imports and blends fertiliser, for example at Athi River) and light as an energy source in

photosynthesis, distinguishing light from a catalyst. The class adds new questions to the DQB and revises the cumulative particle model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Catalysts Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes as Catalysts (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the anchor: both beakers had the same chemicals, and the cross vanished at 35 s (Beaker B, 50 C) and 270 s (Beaker A, 10 C). They list the factors found so far (temperature, concentration, surface area) and what each changes about collisions. Then Mwalimu Achieng shows four measuring cylinders, each with 10 cm³ of 3% hydrogen peroxide and a few drops of detergent, at the same room temperature, labelled: no additive, sand, boiled liver cube, fresh liver cube (each cube 1 cm³). Individually in their science notebooks, learners predict the foam height after 30 s for each cylinder and write a reason, then compare with a partner. They also write: Can a reaction speed up if temperature, concentration and piece size do not change?	Say: 'Last week we found three ways to change how many collisions are effective each second. Turn to your notebook: which factors have we found, and what does each do to the particles?' Take answers from 3 learners by cold-call (include Wanjiru and Kiprono). Hold up the four cylinders: 'The same peroxide, the same temperature, the same amount. Only the thing I drop in is different. Predict the foam height after 30 seconds for each, and give a reason.' WAIT TIME 10-15 seconds in silence while they write. Then say: 'Turn to your partner for 1 minute and compare.' Cold-call 4 pairs for different predictions and write them on the board without judging. Ask: 'Which of the three factors could explain a difference here? If none can, what does that tell us?' Wait 10 seconds. Link to the phenomenon: 'If a fourth factor exists, it should help explain the Disappearing Cross too.'	Predict-Observe-Explain (prediction before evidence) combined with eliciting prior knowledge. Writing a prediction with a reason commits learners to a model, so that surprise at the fresh liver result creates cognitive conflict and motivates revising collision theory. Asking whether any known factor explains the difference makes learners test the existing model against a new case.	Check notebooks for: (1) a prediction for each of the four cylinders with a reason; (2) correct recall of temperature, concentration and surface area with their effect on collisions; (3) whether learners attribute differences to a known factor (misconception: the liver is hotter or 'adds more peroxide'). Tally on the board how many predict that fresh liver differs from boiled liver. Note learners who say sand will act like liver, and revisit them in the Observe phase.
Observe Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes as	Groups of 4 observe the demonstration, each member with an assigned role (reader, timer, foam measurer, recorder). Mwalimu Achieng adds each additive to its cylinder at the same moment and starts the clock. At 30 s groups measure foam height with a ruler and record: no additive = 0 cm; sand = 0 cm; boiled liver cube = 0.5 cm; fresh liver cube = 9	Before starting say: 'Goggles on. Hydrogen peroxide can irritate your eyes, so only I pour it.' Assign roles: 'Reader, timer, measurer, recorder. Hands up when you know your role.' Say: 'Watch all four cylinders at once. Measure the foam height at exactly 30 seconds and record in centimetres.' Circulate and ask each group: 'Which variable did we	Structured data collection with variable identification (fair-test design from Lesson 2). Contrasting cases (sand, boiled liver, fresh liver) isolate the cause: not just any solid, and not liver whose enzyme has been destroyed by boiling. Reusing the liver gives direct evidence that it is not used up, building the defining feature of a catalyst from	Observe whether groups: (1) record 0, 0, 0.5 and 9 cm correctly with units; (2) correctly name independent, dependent and at least four control variables; (3) use the same time for every cylinder; (4) wear goggles and keep the tray under the cylinders. Ask a group member to explain why the boiled liver result differs from fresh liver (look for: boiling damages the

Catalysts Practice Source: CK-12 Search ARES for similar readings	cm. They identify the independent variable (additive), dependent variable (foam height after 30 s) and control variables (10 cm³ of 3% hydrogen peroxide, detergent, temperature, 1 cm³ cube size, time). They test with a glowing splint held by the teacher that gas given off is oxygen: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$. Groups then lift out the fresh liver cube with forceps, rinse and weigh or look at it, and see it is still there and unchanged in size, then add it to fresh peroxide where it again produces foam. They record these observations in a table and in a sketch with labels.	change? Which did we keep the same? Why is sand in the set-up?' (to show that any solid does not work, so it is not just the solid). Ask: 'Why do we have a boiled liver cube?' WAIT TIME 10-15 seconds. Then lift out the fresh liver cube: 'Is the liver still here? Is it smaller? What does that tell us about whether it was used up?' Cold-call 3 learners (include Amina and Otieno). Add the used cube to fresh peroxide: 'Predict, then watch.' Link to the phenomenon: 'Nothing about the temperature, concentration or size of the peroxide changed. Yet the rate changed enormously.'	observation.	enzyme). Learners who say liver 'reacted and disappeared' need to be redirected to the evidence from the retrieved cube.
Explain Phase  VIDEO: Catalysts Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Enzymes as Catalysts (Flexbook) Source: CK-12 Search ARES for similar readings	Learners construct a written explanation answering: Why did the fresh liver cube make 9 cm of foam while sand made 0 cm and boiled liver made 0.5 cm, with the same temperature, concentration and amount of peroxide? Mwalimu Achieng introduces the terms catalyst, enzyme, catalase and activation energy. Learners use a simple energy diagram to show the reaction pathway without a catalyst (high energy hill) and with a catalyst (alternative pathway, lower hill), and link to collision theory: more collisions have enough energy to be effective, so the rate rises, though the particles' speed has not changed. Next, they discuss in groups: (a) the Haber process, where an iron catalyst helps make ammonia, used worldwide for fertiliser (Kenya imports and blends fertiliser, for example blending at Athi River; no Haber plant is claimed there); (b) photosynthesis, where light supplies energy and is not a catalyst; silver bromide in	Say: 'Use the words particles, collisions and activation energy. In two minutes write why fresh liver made 9 cm of foam and sand made none.' WAIT TIME 10-15 seconds before anyone speaks, then a pair share. Cold-call 3 learners to read explanations; ask others: 'Do you agree? What would you add?' Introduce terms: 'Activation energy is the minimum energy a collision needs to react. A catalyst gives the reaction an alternative pathway with lower activation energy, so more collisions are effective, and the catalyst is not used up. An enzyme is a biological catalyst; the enzyme in liver is catalase.' Draw both pathways and ask: 'Did the catalyst make the particles hotter? Did it add more particles?' (No.) Correct misconceptions: 'A catalyst does not make more product; both cylinders would end with the same amount of oxygen, only the time differs.' For light: 'Light supplies energy; it is not a catalyst. Leaves in Kericho tea fields use light for	Claim-Evidence-Reasoning with a visual model (energy pathway diagram). Learners must cite the 0, 0, 0.5 and 9 cm data as evidence and use activation energy as reasoning. Compare-and-contrast of two ways to increase effective collisions (more energy by heating versus a lower energy requirement by a catalyst) strengthens the unified collision-theory idea. Applying the idea to the Haber process and photosynthesis transfers it to new contexts and separates catalysts from light.	Look for written explanations that: (1) cite the data (9 cm versus 0 cm and 0.5 cm); (2) state that the catalyst gives an alternative pathway with lower activation energy; (3) say more collisions are effective; (4) say the catalyst is not used up and does not change the final amount of product; (5) state that light is an energy source, not a catalyst. Misconceptions to track: 'catalyst adds energy', 'catalyst is a reactant', 'boiled liver still works a little because it is a catalyst' (it is a damaged enzyme). Use a quick thumbs-up/down check: 'A catalyst is used up in the reaction.' Ask 4 learners to justify their response.

	photographic film changes in light. They return to the Disappearing Cross and decide what a catalyst would change in Beaker A. They also connect to digestion: the gut uses enzymes so that food breaks down quickly at body temperature.	photosynthesis.' Discuss Haber: 'Why would industry want an iron catalyst?' Cold-call 3 groups. Return to the anchor: 'Explain to a partner in one sentence how temperature and a catalyst each speed up a reaction but in different ways.' (Temperature gives particles more energy; a catalyst lowers the energy needed.) Add: 'In the gut, catalase-like enzymes work at body temperature, so a lower activation energy matters.'		
Driving Question Board (DQB) Creation  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes as Catalysts Practice Source: CK-12  Search ARES for similar readings	Groups return to the DQB on the back wall. They (1) place an answer card on questions from Lesson 1 that can now be partly answered (for example, whether something other than temperature can change the speed); (2) write new questions on sticky notes arising from today, such as: Can a catalyst help Beaker A? Why does boiling stop the liver working? Do catalysts work in cars? How do enzymes in the stomach work at body temperature? Does the Haber process need heat as well? Is light needed for all reactions? Learners place notes under headings: Temperature, Concentration, Surface area, Catalysts, Light, and Other. Kiprono adds a question about whether the enzymes in fermenting uji behave like catalase; Amina adds one about whether anything slows rusting of roofing sheets in Mombasa. These are flagged for Lesson 7.	Say: 'Go to the DQB. Find one question we can now answer with evidence and one new question that today raised.' Allow 3 minutes. Cold-call 4 groups to read one card and one new question aloud. Ask: 'What evidence do we have for that answer? Is it complete?' Wait 10 seconds. Sort questions with the class: 'Which can we answer from today's data? Which need Lesson 7 or 8?' Praise specificity. Challenge vague questions: 'Can you make it testable? What would you change and what would you measure?' Link to the driving question: 'How does what we learned today help make life in Kenya safer, healthier and more productive?' Mention enzyme action in digestion, fertiliser supply and photosynthesis in crops.	Driving Question Board as a public record of growing understanding. Linking new questions to evidence shows how investigation produces further questions, and sorting them into factors keeps the focus on the central question, which keeps learners accountable for tracing each answer back to data.	Check that: (1) answer cards cite evidence from today (such as 9 cm versus 0 cm) and not opinion; (2) new questions are about catalysts, light or energy in reactions; (3) at least one question from each group is testable; (4) learners can say which questions are not yet answerable. Record names of learners whose questions show deeper reasoning (for example asking about enzymes and temperature) for extension in Lesson 7.
Model Building Phase  VIDEO: Factors affecting reaction rates	Individually, learners revise their cumulative particle model for the Disappearing Cross. Their model page from earlier lessons shows particles of thiosulfate and HCl, with changes for temperature,	Say: 'Open to your model page. Add one new idea today: what does a catalyst do to the energy needed for effective collisions? Include a labelled diagram.' Circulate for 5 minutes and ask:	Cumulative model revision with peer critique (glow and grow). Adding a new element to a persisting model shows how new evidence refines an explanation, and drawing both the energy	Model rubric: (a) energy diagram has two labelled pathways with a lower activation energy for the catalyst; (b) states that the catalyst is not used up; (c) explains that more collisions are effective; (d)

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Enzymes and Activation Energy - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	concentration and surface area. Today they add a fourth component: a catalyst, drawn as an alternative energy pathway with lower activation energy, and a note that light can supply energy in some reactions. They draw a labelled sketch of an energy diagram for the hydrogen peroxide reaction with and without catalase, and a particle sketch showing more effective collisions with the catalyst. Then they complete a statement: 'In the Disappearing Cross, the 50 C beaker was faster than the 10 C beaker (35 s versus 270 s) because ... A catalyst would speed a reaction in a different way by ...'. They finish with a short written exit ticket.	'Where did you show the lower activation energy? Is the catalyst in the products?' WAIT TIME 10-15 seconds after each question. Choose 2 models to show (with permission), one with a clear energy diagram, and ask the class to give a glow (what is strong) and a grow (what to add). Cold-call 3 learners: 'Finish this: the warm beaker was 7.7 times faster than the cold beaker because ...' and then 'A catalyst works differently because ...'. Hand out the exit ticket: 'In two sentences, explain why fresh liver made 9 cm of foam and boiled liver made 0.5 cm. Name the type of molecule in liver.' Close: 'Next lesson we apply all four factors to Kenyan life: enzymes in the gut, fermenting uji, rusting and preserving fish.'	diagram and particle picture links the symbolic level to the particle level. Contrasting temperature with catalyst in the same sentence consolidates that there are two different routes to more effective collisions.	light shown as an energy source; (e) correct link back to the Disappearing Cross with 35 s and 270 s. Exit ticket: success means naming the enzyme (catalase/protein) and explaining that boiling damaged it. Learners who write that the catalyst 'gives energy' or 'changes the products' need a quick conference at the start of Lesson 7.
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D. TEACHER REFLECTION

Did learners notice that the rate rose without any change in temperature, concentration or particle size, and did the surprise lead them to revise the model rather than dismiss the result? Which learners said the liver was 'used up', and did retrieving the cube change their thinking? Were learners able to separate how a catalyst works (lower activation energy) from how heating works (more energy), and did anyone confuse light with a catalyst? Which DQB questions need Lesson 7 (optimum temperature and pH for enzymes, boiling denatures catalase) and Lesson 8? Check the safety routine: goggles worn, peroxide handled only by the teacher, hands washed after liver.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In 30 s, 10 cm ³ of 3% hydrogen peroxide with a few drops of detergent produced foam of 0 cm with no additive, 0 cm with sand, 0.5 cm with a boiled liver cube and 9 cm with a fresh liver cube (1 cm ³ each). The fresh liver cube was still there afterwards and worked again in new peroxide. Temperature, concentration and amount were the same in every cylinder.
What did I learn?	A catalyst speeds up a reaction without being used up and does not change the final amount of product. The enzyme catalase in fresh liver is a biological catalyst; boiling damages it. Light is an energy source for some reactions (photosynthesis) and is not a catalyst. Catalysts are used in industry, such as the iron catalyst in the Haber process for ammonia.
How does this explain the phenomenon?	A catalyst gives the reaction an alternative pathway with a lower activation energy, so a larger fraction of collisions have enough energy to be effective collisions, and the rate rises without changing temperature, concentration or size. This adds to the Disappearing Cross story: warm Beaker B (35 s) beat cold Beaker A (270 s) because heating gives particles more energy, while a

	catalyst lowers the energy needed.
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LESSON 7 (80 minutes (2 x 40-minute lessons)): Applying the Model to Biological, Chemical and Physical Processes in Kenya

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the collision-theory model built in Lessons 3 to 6 to real Kenyan systems. They see that several factors act together, that enzymes have optimum conditions, and that preservation works by slowing reactions.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain factors that influence the rate of reactions in biological, chemical and physical processes; (b) describe why enzymes such as amylase (near 37 C, pH 7) and pepsin (near pH 2) work best at optimum conditions; (c) explain how cooling, drying and salting slow reactions that spoil food; (d) appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life.
Skills	Analysing and interpreting data (NGSS SEP 4); Constructing explanations (SEP 6); Developing and using models (SEP 2); Engaging in argument from evidence (SEP 7); Obtaining, evaluating and communicating information (SEP 8).
Attitudes	Respect for others' opinions during group discussion; responsibility when using food and materials; appreciation of how science improves food safety and livelihoods in Kenya. Core competencies: Communication and collaboration; Critical thinking and problem solving. PCIs: Safety and security; Socio-economic issues (food preservation).
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lessons 3 to 6 gave the model that rate depends on how many effective collisions happen per second: temperature, concentration, surface area, and catalysts and light. Lesson 7 moves the model out of the beaker into Kenyan life, such as the gut, uji, roofing sheets and fish at Lake Victoria. It also revisits the Lesson 1 DQB questions, including why Beaker B (35 s) was faster than Beaker A (270 s), and prepares for the Lesson 8 final model.
Safety Notes	Use only food-grade materials, and do not taste anything made in the laboratory. Wash hands before and after handling uji, salt and fish pieces. Wear goggles if any iron nails and salt solution are used. Keep the tutorial paper-based or use samples prepared in advance. Wipe up spills at once. No thiosulfate-HCl reaction is repeated, so SO ₂ is not a risk in this lesson.

B. LESSON OVERVIEW

Mwalimu Achieng reopens the Disappearing Cross. Beaker B, at 50 C, lost its cross at 35 s, and Beaker A, at 10 C, lost it at 270 s, so B was 7.7 times faster with identical chemicals. Learners predict what happens to reactions in real Kenyan systems when conditions change: uji left in a warm kitchen at 30 C versus a cool place at 15 C, enzymes in the gut, iron roofing sheets in Mombasa, and fish at Lake Victoria. They analyse reference values (amylase near 37 C and pH 7, pepsin near pH 2, uji 24 h at 30 C versus 48 h at 15 C, fridge 4 C) and revisit the Lesson 3 temperature table as evidence for cooling food. They explain why enzymes have an optimum, how cooling, drying and salting slow spoilage, and that several factors act together. They then review the Lesson 1 DQB questions and revise the class model. The factors change the speed of a reaction, not the final amount made.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at the Beaker A /	Say: 'Last time we saw the cross	Predict-Observe-Explain (POE)	Look for: (1) predictions that name

 VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes and Activation Energy - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Beaker B record in their science notebooks (10 C, 270 s; 50 C, 35 s). Mwalimu Achieng shows the scenario cards: (1) uji in a warm kitchen at 30 C versus 15 C, (2) a fridge at 4 C holding milk and meat, (3) fresh fish drying in the sun at Lake Victoria. In pairs, learners predict in their notebooks: 'Which will spoil or react faster, and why?' They draw one particle sketch showing what changes in the warm and cool cases. Wanjiru's earlier 'why' question is read aloud to link back to the beaker.	vanish at 35 seconds in warm Beaker B and at 270 seconds in cold Beaker A. Today we ask whether the same idea explains uji, a fridge and fish.' Show the three cards and give WAIT TIME of 10-15 seconds of silent thinking before any talk. Then say: 'Write your prediction and one particle sketch first.' Cold-call 4 learners (Wanjiru, Otieno, Amina, Kiprono, or other learners using the same role names) to share predictions without correcting them. Say: 'Do not tell me if it is right yet. Tell me what you think the particles are doing.' Record every prediction on chart paper for revisiting at the end.	with a particle sketch: committing to a prediction and a mechanism before seeing evidence exposes learners' current model of collisions and creates a clear reason to test it against the data.	a direction (warmer spoils faster) and not just 'it changes'; (2) sketches showing particles moving faster or colliding more in the warm case; (3) learners who use the words 'particles' or 'collisions'. Tally how many of the 4 cold-called learners link their reasoning to temperature. Note learners who say a fridge 'stops' reactions, as a misconception to address in Explain.
Observe Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes and activation energy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups of 4 receive a data pack with the Lesson 7 reference values. Part A (temperature and food): uji ferments in about 24 h at 30 C but about 48 h at 15 C; a fridge is at 4 C. Learners reopen the Lesson 3 temperature table (10 C = 270 s, rate $3.7 \times 10^{-3} \text{ s}^{-1}$; 25 C = 110 s, rate 9.1; 40 C = 52 s, rate 19.2; 55 C = 26 s, rate 38.5) and compare how rate changes with temperature. Part B (enzymes): they read a simple set of approximate classroom values: amylase works best near 37 C and pH 7, and pepsin in the stomach works best near pH 2. They draw a sketch graph of rate (y-axis) against temperature or pH (x-axis) for amylase, showing rate rising to a peak near 37 C and then falling. Part C (preservation): they examine sample cards or pictures of dried and salted fish from Lake Victoria, of sun-drying, and of rusting iron roofing sheets in Mombasa (Amina's example), and list what condition is being	Circulate and ask: 'Which row of the Lesson 3 table is closest to your kitchen? What happens to the time when temperature drops?' Ask groups: 'The uji takes 24 hours at 30 C and 48 hours at 15 C. How many times longer is that, and what does that tell you about the rate?' (Expect: 2 times longer, so the rate is about half.) Say: 'Use evidence from the table, not just opinion.' Prompt: 'Amylase is a biological catalyst, and a catalyst speeds up a reaction. So why does its rate fall if we keep heating above 37 C?' Give WAIT TIME of 10-15 seconds, then cold-call 3 groups. Say: 'Do not give me the answer yet. Point to the part of your graph where the rate changes direction.' Remind groups that the final amount of product is not the focus; the speed is.	Data analysis with a sketch graph and gallery comparison: converting reference values into rates and plotting a peak shape lets learners notice patterns (steady rise for ordinary reactions, a peak for enzymes) and stops them from assuming 'hotter is always faster'.	Check that learners (1) calculate the 24 h versus 48 h comparison correctly (2 times, rate about half); (2) quote the 10 C and 25 C rows of the Lesson 3 table to support 'cooler means slower'; (3) draw the enzyme graph with a peak and not a straight line; (4) list the condition changed for each preservation case (cooling, drying, salting). Use a checklist for each group. Ask learners who draw a straight line to point to the 37 C value on the graph.

	changed in each case.			
Explain Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Reaction Rate (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Each group builds a written explanation for one case using the collision-theory model: (a) uji at 30 C versus 15 C; (b) why a fridge at 4 C slows spoilage; (c) why amylase and pepsin have different optimum conditions; (d) why drying and salting fish stop spoilage; (e) why iron roofing sheets rust faster in Mombasa than in a dry area (more moisture and salt, and a warm climate). Learners use a claim-evidence-reasoning frame: Claim, Evidence (a number or table row), Reasoning (particles, effective collisions, activation energy, kinetic energy). They also decide which of the four factors (temperature, concentration, surface area, catalyst) is acting in each case and note when several act together, for example in the gut: body temperature of 37 C, chewed food (larger surface area), enzymes (catalysts) and a suitable pH. Kiprono links this to uji and tea from Kericho. Groups present in 90 seconds each.	Say: 'Every explanation must include one number and the word particles.' Model once: 'At 15 C the uji particles have less kinetic energy, so fewer collisions have enough energy to react. This is fewer effective collisions per second, so 48 hours.' Then ask: 'Which factors are acting together in your case?' Introduce the idea that an enzyme has a shape that works best at its optimum pH and temperature; outside it, the enzyme works poorly or stops (too hot changes its shape, so it no longer catalyses). Clarify the misconception: 'A fridge slows the reaction; it does not stop it, and drying and salting remove water that microorganisms and reactions need.' Cold-call 5 learners to restate another group's reasoning in their own words. Use WAIT TIME of 10-15 seconds after each question. Do not re-teach the individual factors, only link them.	Claim-Evidence-Reasoning (CER) with group presentations and peer restatement: tying every claim to a data point and a particle-level mechanism builds the habit of argument from evidence and shows that one model explains many contexts.	Observable indicators: (1) the claim states a direction and a comparison; (2) evidence quotes a value exactly (24 h / 48 h, 37 C, pH 2, or a row of the temperature table); (3) reasoning uses 'effective collisions', 'kinetic energy' or 'activation energy' correctly; (4) learners identify at least two factors acting together in the gut. Score each group 0-3 on claim, evidence, reasoning. Listen for the misconception 'the fridge stops the reaction' and for 'more heat always speeds up an enzyme'.
Driving Question Board (DQB) Creation  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Chemical Reaction Rate Practice <i>Related topic: not</i>	At the back wall, learners and Mwalimu Achieng review the Lesson 1 DQB questions, including Wanjiru's 'why did Beaker B finish so much faster?' and the questions about mango rotting, ugali cooking, uji and rusting. In groups, learners decide which questions can now be answered, write the answer on a sticky note with the evidence, and move the note beside the question. They add new questions, for example 'Why do some foods keep longer when salted?' and 'Why does a fever change how body reactions work?' Unanswered	Say: 'Let us go back to the first questions we put up. Which of them can we answer with evidence today?' Hand out sticky notes and ask each group to move at least one answer to the board. Give WAIT TIME of 10-15 seconds, then cold-call 4 learners to read a question and its answer aloud. Ask: 'What evidence supports that answer? Which lesson gave it to us?' Ask Amina and Kiprono to link the rusting and uji questions to their examples. Circle any question that still has no evidence and say: 'We will use these in Lesson 8.'	Driving Question Board revisit: matching earlier questions to the evidence gathered makes learning progress visible and shows learners that the whole sequence is moving toward answering the driving question.	Check that each group has moved at least one note and attached evidence, not just a statement. Look for answers that cite a lesson (for example 'Lesson 3 table: 10 C = 270 s') and for new questions that show deeper thinking about optimum conditions or preservation. Count the questions answered and the questions left open.

<i>an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	questions are marked for Lesson 8.			
Model Building Phase  VIDEO: Catalysts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Enzymes and Activation Energy - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners return to their individual particle model of the Disappearing Cross and revise it. They add a section titled 'Real systems in Kenya' with at least three annotated examples (uji, enzymes in the gut, preserved fish or a fridge) showing which factor changes effective collisions per second. They add a note on optimum conditions: an enzyme has a best temperature and pH (amylase near 37 C and pH 7; pepsin near pH 2). They add a note that cooling, drying and salting slow reactions, and that the factors change the speed, not the final amount of product made. They then compare their first prediction with their revised model and write one sentence: 'I used to think ..., now I think ... because ...'.	Say: 'Open the model you started in Lesson 1. Add what you can now explain, and cross out any idea the evidence does not support.' Display the sentence frame: 'Because ____ (factor), particles ____, so ____ effective collisions per second, so the rate is ____.' Walk around and prompt: 'Which factor is acting here? Where is the evidence?' Pair learners for a 2-minute peer review and give WAIT TIME of 10-15 seconds before they respond. Cold-call 3 learners to read their revision sentence. Return to the beaker and ask: 'If the final model must explain both Beaker B (35 s) and the fridge, what idea do they share?' Expect: more or fewer effective collisions per second because particles have more or less kinetic energy.	Cumulative model revision with before-and-after comparison: updating the same model each lesson and stating what changed makes learners' reasoning visible and helps them see one mechanism explaining the beaker, the gut and the kitchen.	Look for models that (1) include at least three Kenyan examples linked to a named factor; (2) use 'effective collisions' and 'activation energy' correctly; (3) show an optimum for enzymes and not a simple rise; (4) state that final amount is unchanged. Collect the 'I used to think ... now I think ...' sentences as an exit ticket and sort them into secure, developing and misconception piles to plan Lesson 8.

D. TEACHER REFLECTION

After the lesson, consider: Did learners explain the enzyme peak with particles and shape, or did they just say 'too hot kills it'? Which learners still think a fridge stops reactions completely? Did groups quote a number from the Lesson 3 table and the 24 h / 48 h values? Were at least two factors linked together in the gut example? Which DQB questions remain unanswered, and which will be used in the Lesson 8 assessment? Note any learners who need support with sketching graphs or with using 'effective collisions' accurately.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We looked at the Beaker A / Beaker B result (10 C = 270 s; 50 C = 35 s) and compared it with Kenyan examples: uji fermenting in about 24 h at 30 C but about 48 h at 15 C, a fridge at 4 C, amylase working best near 37 C and pH 7, pepsin near pH 2, dried and salted fish, and rusting roofing sheets in Mombasa.
What did I learn?	Several factors act together in real systems. Enzymes (biological catalysts) have optimum temperature and pH. Cooling, drying

	and salting slow the reactions that spoil food. The factors change the speed of a reaction, not the final amount made.
How does this explain the phenomenon?	Because cooler conditions give particles less kinetic energy, fewer collisions have enough energy (activation energy) to react, so there are fewer effective collisions per second and the rate is lower. Enzymes work best when their shape suits the temperature and pH. Drying and salting remove the water needed for spoilage reactions, so food keeps longer.

LESSON 8 (80 minutes (2 x 40-minute lessons)): Model Building, Assessment and Community Connection: The Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To let learners put the cumulative collision-theory model to use: to explain the anchor demonstration in full, predict and explain an unfamiliar Kenyan scenario, and choose which factor to change to solve a real problem.
Knowledge	By the end of the lesson, the learner should be able to: a) describe rate of chemical reactions (rate = $1/t$ for the cross method; cm^3/s for the gas method); c) explain factors that influence the rate of reactions in biological, chemical and physical processes (temperature, concentration, surface area, catalyst, light) using effective collisions and activation energy; d) appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life. Final explanation of the anchor: Beaker B (50 C) cross vanished at 35 s and Beaker A (10 C) at 270 s, so B was 7.7 times faster (rates 28.6 and $3.7 \times 10^{-3} \text{ s}^{-1}$). Same chemicals, same amounts, only temperature differed. Both beakers end with the same amount of sulfur; factors change the speed, not the final amount made.
Skills	Constructing a scientific explanation (claim, evidence, reasoning); using data from earlier tables as evidence (temperature table 25 C = 110 s, 40 C = 52 s; concentration table 0.15 mol/dm^3 = 110 s and 0.06 mol/dm^3 = 275 s; surface area table initial rates 0.40, 0.93 and 2.07 cm^3/s); comparing the Lesson 1 model with the final model; predicting from a model; choosing and justifying a factor to change in a real problem; communicating a message to the community.
Attitudes	Responsibility and respect (KICD values): listening to and building on classmates' reasoning during the gallery walk and critique. Appreciation of how controlling reaction rate makes life in Kenya safer, healthier and more productive (food preservation, digestion, cooking, protecting roofing). Curiosity: valuing the questions that remain open on the DQB. Safety and security (PCI): safe handling of any materials and ventilation.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	This is the final lesson. In Lesson 1 learners saw the cross vanish at 35 s in warm Beaker B and 270 s in cold Beaker A and could only say 'heat makes it faster'. Lessons 2 to 7 gave them rate = $1/t$ and fair tests (L2), temperature and first collision theory (L3), concentration (L4), surface area (L5), catalysts and light (L6) and optimum conditions in Kenyan systems (L7). Now they bring everything together in a Final Explanation of the anchor, close out the DQB, compare their first and final models, and show they can use the model on a new problem.
Safety Notes	No new chemical work is required. If Beakers A and B are re-shown, use the same small quantities (10 cm^3 of 1.0 mol/dm^3 HCl, 50 cm^3 of 0.15 mol/dm^3 sodium thiosulfate), a ventilated room, goggles, and rinse beakers straight after the cross vanishes because SO_2 is an irritant. Prefer showing the L1 notebook records or a replay instead of repeating the demo. For the transfer task use only paper-based or safe household examples (no heating of oil, no open flames beyond what is normal in a school kitchen). Keep walkways clear for the gallery walk.

B. LESSON OVERVIEW

Mwalimu Achieng opens by asking the class to predict, in their science notebooks, what they now think a particle-level picture of Beakers A and B looks like and how they would solve an unfamiliar Kenyan problem: a trader in Kisumu whose fresh fish spoils in hours, and a farmer in Kericho choosing how to speed up composting of tea waste. Learners revisit the anchor evidence (35 s vs 270 s, 7.7 times faster, rates 28.6 and $3.7 \times 10^{-3} \text{ s}^{-1}$) and write a full Final Explanation using claim, evidence and reasoning, citing one row from the temperature, concentration or surface area tables. They apply the same model to the new scenario, with no new numbers,

deciding which factor to change and why. Wanjiru, Otieno, Amina and Kiprono's DQB questions are sorted into answered, partly answered and still open. Learners then compare their Lesson 1 model with a final class model, noting what changed, and finish with a short community message about controlling reaction rates and a link to careers such as KEMRI biochemists.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their notebooks to the Lesson 1 page on which they recorded the anchor observation (Beaker B cross gone at 35 s, Beaker A at 270 s). Individually (3 minutes) they sketch a particle-level picture of each beaker and write a prediction for a NEW problem with no numbers: 'A fish trader in Kisumu wants her tilapia to stay fresh for a whole day without a fridge. Which factor would you change, in which direction, and what will happen to the rate of the spoiling reactions?' They then pair up with a partner and compare predictions, noting where they used words like particles, collisions and energy.	Mwalimu Achieng says: 'Open to your Lesson 1 page. Two beakers, same chemicals, same amounts. Draw what you now think the particles are doing in A and in B. Then tell me how you would slow spoilage of fish at Lake Victoria.' She gives WAIT TIME of 10-15 seconds of silence before pairs talk. She circulates and cold-calls 4 learners (including Wanjiru and Kiprono) to read one prediction aloud without correction, saying: 'Thank you. I am not telling you if it is right yet. We will test it with evidence.' She records 3 predictions on the board under 'Our predictions before the final explanation'.	Predict-before-instruction with an individual-then-pair 'Think-Pair-Share': learners commit to a prediction first, so they can later see for themselves how their thinking has changed. It also activates their prior model from Lessons 1 to 7 so the Explain phase builds on their own ideas rather than the teacher's.	Look for: (1) sketches with particles drawn moving faster or more crowded in B, not just a label 'hot'; (2) use of the words 'particles' and 'collide' in at least half of the predictions; (3) any prediction that names a factor (temperature, drying, salting, surface area) for the fish. Tally on a class list how many learners link their fish prediction to collisions. Learners who give only 'because it is cold' receive a prompt in the Explain phase: 'What are the particles doing?'
Observe Phase  VIDEO: Factors affecting reaction rates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Factors Affecting Reaction Rate Practice Source: CK-12  Search ARES for similar readings	Evidence review in groups of four. Each group receives a printed 'Evidence Pack' with only data already collected: the anchor result (A: 270 s, rate $3.7 \times 10^{-3} \text{ s}^{-1}$; B: 35 s, rate $28.6 \times 10^{-3} \text{ s}^{-1}$; 7.7 times faster; 235 s difference), the temperature table (10 C = 270 s, 25 C = 110 s, 40 C = 52 s, 55 C = 26 s), the concentration table (0.15 to 0.06 mol/dm ³ giving 110 to 275 s, rate/concentration about 60), the surface area initial rates (0.40, 0.93, 2.07 cm ³ /s; maximum 120 cm ³), the catalyst foam heights (0, 0, 0.5, 9 cm) and the Lesson 7 values (amylase near 37 C and pH 7; pepsin near pH 2; uji 24 h at 30	Mwalimu Achieng says: 'You are detectives. Find ONE row of data that best explains why Beaker B was faster, and be ready to defend it.' She hands out packs and sets a 10-minute timer. She asks groups: 'In the anchor, what was kept the same? What was different?' WAIT TIME 10-15 seconds after each question. She cold-calls 3 groups to name their chosen row and says: 'Why did you pick that one over the others?' She reminds them to use only the given numbers and not invent any new ones. She points to the DQB and says: 'Which of these numbers answers one of our questions?'	Evidence sorting with a 'same / different' highlight: learners identify control and independent variables across all earlier investigations, which shows that the same particle idea runs through every factor. Looking across four datasets lets learners see the pattern (more effective collisions per second means a higher rate) instead of seeing seven separate facts.	Observable indicators: groups correctly identify temperature as the only difference between Beakers A and B; at least one row quoted accurately (for example 40 C = 52 s or 0.06 mol/dm ³ = 275 s); groups can state that the amount of sulfur at the end is the same. Use a quick checklist per group. If a group misquotes a value, send them back to the pack. Reteach later if two or more groups confuse rate with time.

	C vs 48 h at 15 C; fridge 4 C). Groups highlight patterns in two colours: 'what changed' and 'what stayed the same'. Each group notes that both beakers produce the same final amount of sulfur and that the factors changed only how fast. They also look at the new scenario (fish at Lake Victoria dried, salted or kept cold; tea waste composted in Kericho) and list which factors are present in each.	Where groups mix up rate and time she asks: 'Is a bigger rate a longer time or a shorter time?' She also checks that no group claims a factor changes the final amount of product.		
Explain Phase  VIDEO: Collision theory and the Maxwell-Boltzmann distribution <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rates (PLIX) Source: CK-12  Search ARES for similar readings	Part 1 (individual, 10 minutes): Each learner writes a FINAL EXPLANATION of the anchor in their notebook using the frame: CLAIM (why was Beaker B faster?), EVIDENCE (35 s vs 270 s, 7.7 times faster, rates 28.6 vs $3.7 \times 10^{-3} \text{ s}^{-1}$, plus one row from the temperature, concentration or surface area table), REASONING (particles, kinetic energy, collisions, effective collisions, activation energy, and that the same amount of sulfur forms in the end). Part 2 (groups, 10 minutes): Learners then use the model on the new scenario: 'In Kisumu, a trader dries and salts fish; in Kericho, Kiprono's family wants tea-leaf waste to compost faster. Which factor would you change and why?' They write a reasoned recommendation, for example: cooling or drying fish lowers the rate of spoilage reactions, as shown by the temperature table (lower temperature gives fewer effective collisions per second); chopping the tea waste increases surface area and warmth increases the rate, like the powder at 2.07 cm³/s versus large chips at 0.40 cm³/s. Part 3: Pairs swap explanations and use a peer	Mwalimu Achieng models the frame with a sentence starter on the board: 'Beaker B was faster because ... The evidence is ... This happens because ...'. She says: 'A good explanation uses the words particles, energy, collide and effective. Which of these will you use first?' WAIT TIME 10-15 seconds. She asks Otieno, Amina, Kiprono and Wanjiru (4 cold-calls) to share one sentence from their reasoning, linking to their own earlier contributions: Otieno's particle idea (L1), Amina's rusting roof, Kiprono's uji and tea, Wanjiru's first 'why'. She probes: 'Does higher temperature only mean more collisions, or also something about energy?' and 'Did the factors change how much sulfur was made, or how fast?' She corrects misconceptions: 'frequency of collisions' is not the same as energy of collisions; a catalyst is not used up; light is a source of energy and not a catalyst. She checks that the new scenario uses no new numbers: 'Use only the data in your pack to justify your choice.' She selects 2 strong explanations to read aloud (with permission) and annotates them on the board.	Claim-Evidence-Reasoning (CER) scientific argumentation, followed by transfer to an unfamiliar scenario. CER forces learners to connect observed data to the particle model, and transfer shows whether they can use the model as a tool rather than recite it. Peer review with 'star and step' builds a shared standard for a good explanation.	Use a 4-point rubric on the notebook explanation: (1) correct claim naming temperature; (2) accurate evidence citing 35 s and 270 s plus one table row; (3) reasoning including kinetic energy, effective collisions and activation energy; (4) recognises that the final amount of sulfur is unchanged. For the new scenario look for a chosen factor, a direction (increase or decrease), and a reason using collisions. A learner at level 2 or below receives a targeted question at the end of the lesson; note names for follow-up. Collect 6 notebooks as exit evidence.

	checklist to give one 'star' and one 'step'.			
Driving Question Board (DQB) Creation  VIDEO: The Arrhenius equation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Reaction Rate (Flexbook) Source: CK-12  Search ARES for similar readings	At the back wall, learners return to the DQB that Mwalimu Achieng has kept since Lesson 1. Working in their groups, each takes 3 to 4 questions (for example Wanjiru's 'Why was B faster if the chemicals are the same?', Amina's question about rusting roofing sheets in Mombasa, Kiprono's question about uji and tea, and the L1 DQB questions revisited in L7). They sort each question with sticky notes into three columns: ANSWERED (with the lesson number and evidence), PARTLY ANSWERED, and STILL OPEN. Each learner then adds one new question they would like to investigate next (for example about pressure, which was named in the KICD list but not investigated here, or about how medicines are stored in hot places). Finally, the class returns to the whole driving question and each group completes the sentence: 'We can control the speed of reactions to make life in Kenya safer, healthier or more productive by ...'.	Mwalimu Achieng says: 'Walk to the board. For each question, point to the data that answers it. If you cannot point to data, it is not yet answered.' WAIT TIME 10-15 seconds before groups move. She cold-calls 3 groups to defend a placement: 'Why do you say this is only partly answered?' She asks Amina to explain her roofing-sheet question in terms of rate and conditions (with L7 content only), and Kiprono to link uji at 30 C vs 15 C (24 h vs 48 h) to the model. She says: 'A question that stays open is not a failure; it is where scientists go next.' She records the new questions in a list headed 'Questions for next time'. She checks that every group has linked at least one answered question to a named lesson.	DQB sorting and revisiting: classifying questions as answered, partly answered or open makes learning visible and shows learners the growth in their understanding over eight lessons. Writing a new question models the way scientists extend knowledge and gives ownership of what happens next.	Look for: each placement justified with a specific piece of evidence (for example 'answered: L3 temperature table, 40 C = 52 s'); at least one question per group honestly placed as partly answered or open; new questions that are testable and tied to reaction rate. Record the count of questions moved to ANSWERED as a class indicator. If a group places a question as answered without evidence, ask: 'Which row shows that?'
Model Building Phase  VIDEO: Rate law and reaction order <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners take out the Lesson 1 model they drew (their first idea of why B was faster) and place it beside a blank page. Working in groups they build the FINAL MODEL on a large sheet: particle diagrams of Beakers A and B showing speed and collisions, with labelled effective collisions, activation energy, and a panel for each factor (temperature, concentration, surface area, catalyst, light) showing what changes. They add a note that the	Mwalimu Achieng says: 'Your final model must be able to explain the two beakers AND the fish AND one more case. If it cannot, revise it.' She provides sugar paper, markers and a model checklist (particles, kinetic energy, effective collisions, activation energy, five factors, same final amount). WAIT TIME 10-15 seconds before groups begin drawing. She circulates asking: 'Where in your model does the temperature table row show up?' and 'What does	Cumulative model revision with side-by-side comparison: placing the Lesson 1 model next to the final model makes the change in thinking concrete and builds metacognition. The gallery walk with written comments uses peer feedback, and the community message turns the model into a tool for action, linking science to the driving question.	Observable indicators in the final model: (1) both beakers drawn with particle speed and collisions differing; (2) all five factors represented with a correct direction of effect; (3) a correct note that factors change the rate, not the final amount of sulfur; (4) comparison table with at least two substantive changes from Lesson 1; (5) a community message naming one factor, one direction and one collision-based reason. Score each poster with the 4-level

 EXERCISE: Factors Affecting Reaction Rate Practice Source: CK-12  Search ARES for similar readings	amount of sulfur made at the end is the same in both beakers; only the rate differs. They then complete a comparison table: 'Lesson 1 model' vs 'Final model' with rows for What I thought, What I know now, Evidence, and What I still wonder. In the final 8 minutes, each group chooses one real Kenyan problem (fish preservation at Lake Victoria, uji fermentation, rusting roofing sheets in Mombasa, digestion with amylase, fertiliser production with an iron catalyst in the Haber process, which is used worldwide) and prepares a 30-second community message on which factor to change, and why. Groups do a gallery walk, leaving one comment on two other posters. A short closing talk links these ideas to careers, such as biochemists at KEMRI.	your model predict if the fish is dried AND kept cold?' She cold-calls 4 learners during the gallery walk: 'What changed between your first model and this one?' She gives each group 30 seconds for the community message and says: 'Speak to a farmer or trader, not a scientist. Use one factor and one reason.' In the last 3 minutes she invites Wanjiru to read her first 'why' question from L1 and then answer it in her own words. She closes by asking learners to rate their confidence from 1 to 4 in their notebooks and to write one sentence on how reaction rates matter to their own family.		rubric (exceeds, meets, approaches, below). Note groups at 'approaches' or below for follow-up in the next lesson. Use the 1 to 4 confidence rating as an exit ticket.
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D. TEACHER REFLECTION

Which learners could explain the anchor using effective collisions and activation energy rather than just 'heat makes it faster'? Did anyone still claim that a factor changes how much sulfur forms, or that a catalyst is used up, or confuse rate with time? Which evidence rows did learners choose most often, and did they quote them accurately without adding new numbers? How did Wanjiru, Otieno, Amina and Kiprono's DQB questions help others to see the link between lessons? Which transfer scenario (fish, tea waste, uji, roofing) produced the strongest reasoning, and which needs more support? What would I change about timing, grouping or the Evidence Pack the next time I teach this sub-strand?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the anchor demonstration, the cross vanished at 35 s in warm Beaker B (50 C) and at 270 s in cold Beaker A (10 C): 7.7 times faster, with rates 28.6 and $3.7 \times 10^{-3} \text{ s}^{-1}$. The same chemicals and amounts were used, and both ended with the same amount of sulfur. Add one row of evidence from the temperature, concentration or surface area table.
What did I learn?	Reaction rate depends on how many effective collisions happen per second. Higher temperature, higher concentration, larger surface area and a catalyst all increase the rate; light supplies energy for some reactions. Factors change how fast, not how much, product forms. Optimum conditions exist in biological systems, and cooling, drying and salting slow reactions to preserve food.
How does this explain the phenomenon?	Write your Final Explanation: Beaker B was faster because at 50 C the particles have more kinetic energy, so they collide more often and with more energy, and a larger fraction of collisions have at least the activation energy, making them effective. Cite your

	evidence row. Then explain how you would change one factor to solve a real Kenyan problem (for example preserving fish at Lake Victoria) and how your model has changed since Lesson 1.
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DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.